INCH-POUND

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PERFORMANCE SPECIFICATION
CONTROL UNIT, COMMUNICATION SYSTEM C-11746(V)( )/ARC

This specification is approved for use by the USACECOM, Department of Army , and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a type of communications control unit designated as Control Unit, Communication System C-11746(V)( )/ARC. This performance specification is intended as a form, fit, and functional replacement for the "standard" Control Unit, Communication System C-11746(V)/ARC (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5895

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2.2 Other Government documents, drawings and publications.

PUBLICATIONS

NACSEM-5100	-	Compromising Emanations Laboratory Test Standard, Electromagnetic (Document CONFIDENTIAL Title unclassified).

(The above Publications are classified but are available for review from U.S. Army Communications Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 for those having proper security clearances on file).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein , the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sampling shall be subjected to first article inspection.

3.2 Design, materials, and manufacturing. Unless otherwise specified, the design, materials, and manufacturing process selection is the prerogative of the contractor provided all articles submitted to the government fully meet the operating, interface, interchangeability/ interoperability, and environmental requirements specified in this document.

3.3 Finish. The equipment finish shall contribute to field camouflage through color selection and reflectance of all externally visible components. All parts shall be suitably treated or finished to provide protection from corrosion. Protective coating shall be field restorable.

3.4 Marking. Mark directly or by label to include, contract number, unique serial numbers on each nomenclatured item and circuit card assembly (CCA), date, manufacturer code, part number, and national stock number. Use clear and legible lettering. The selected method of marking shall withstand the specified operating environmental conditions without any evidence of breakage.

3.5 Electrical preconditioning. Prior to submission to first article inspection, each system built shall be electrically preconditioned for 48 hours. The last 24 hours of preconditioning shall be failure free. Failures which occur in test shall be noted and reported, but will not be counted towards the establishment of the equipment MTBF. Duty cycle imposed, vibration, environmental

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test profile and parameters to be measured shall be those included in the contractor's approved reliability test procedure.

3.6 Safety. The equipment shall be physically and electrically safe to operate.

3.7 Operating requirements. Each Communication System C-11746(V)( )/ARC Control Unit shall provide the following functional, operational, and performance capabilities:

3.7.1 Remote control. The unit shall be capable of accommodating two types of remote control. In the sequential mode, transmit selection shall be by means of an external momentary contact switch. In the SP5T mode, transmit selection shall be accomplished by actuating one of five momentary contact switches, one for each radio channel. Operation of one of the switches shall also key the selected radio. The SP5T switch shall have five positions, namely; up, down, left, right, center-in.

3.7.2 Piggy back configuration. The unit shall be capable of being configured in tandem with another unit to provide selection and control of 10 radios at a single station. This configuration shall be accomplished by external connections only. Remote sequential mode control shall be operable in this configuration.

3.7.3 VOX. The unit shall be capable of voice actuated mic keying for the ICS channel only. This feature will be selectable from the front panel HOT MIC/VOX ON/NORM/ICS OFF switch. In addition, VOX threshold control shall be available on the front panel.

3.7.4 Weight. The weight of the completely assembled unit shall weigh no more than 2.6 pounds.

3.7.5 Outline dimensions. The outline dimensions of the completely assembled unit shall be within the following limitations:

Panel/Housing Height - 2.625 inches (6.67 cm)

NOTE: Housing height behind front panel such that dimension .048 min. between top of panel and top of housing and between bottom of panel and bottom of housing.

Panel Width - 5.75 inches (14.60 cm)
Housing Width - 5.0 inches max. (12.77 cm)
Housing Depth - 5.12 inches (11.75 cm) (excluding front panel and rear connectors)

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3.7.6 Securing of parts. Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified environmental and field service conditions.

3.7.7 Front panel. The equipment shall have an integrally illuminated front information panel. The panel shall have white legends against a black background for daylight operation and shall have green primary lighting for night operation, to provide compatibility with night vision goggles. The panel lettering font shall be Alternate Gothic No. 3.

3.7.8 Reliability. The equipment shall have a Mean-Time-Between Failures (MTBF, θ_1) of at least 3,500 hours when subjected to ambient air temperature between -65.2°F to 131°F operating at 100% duty cycle (5 minutes receive and 1 minute transmit) in an airborne environment.

3.7.9 Chromaticity and spectrum. Normal legend illumination shall be ANVIS GREEN A with U' coordinate of .099, V' coordinate of .544 and a radius of .034 on the 1976 UCS chromaticity diagram measured at 0.1 ft. lambert display luminance. The spectral radiance shall be such that the ANVIS RADIANCE (AR) shall not exceed 1.7×10^{-10} AR when illuminated to produce 0.1 ft. lambert (FL) display luminance.

3.7.10 Brightness. The brightness shall be 1.0 ± 0.5 foot-lamberts at nominal panel supply voltage.

3.7.11 Power. Panel lighting shall be operable from the 28 volt dc or 5 volt dc/ac (400 Hz) aircraft source. Green panel lighting equipment shall also be operable from the 115V ac (400 Hz) or 28 volt dc aircraft source.

3.7.12 Panel protection. The CSC shall be designed so that each panel type, i.e., 5 volt, 28 volt, or 115 volt, receives its power over a separate power input pin, both at the input connector J1 and, at the panel connector. This is required to prevent damage to the CSC should it inadvertently be installed in the incorrect aircraft.

The following pins are designated for panel illumination in input connector J1:

Panel illumination voltage	Pin number
5 Volts	R
28 Volts	E
115 Volts	J
Common	U

3.7.13 Alternate front panel red. As specified in the contract or purchase order, units shall have an alternate integrally illuminated front information panel. The panel shall have white

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legends against a black background for daylight operation and shall have red primary lighting for night operation. All lighting shall be deviation-instrument panel lighting red (@.620) and shall be externally controllable. By a change of panel, the unit(s) shall be compatible with either a nominal 28 volt dc or a nominal 5 volt dc/ac (400 Hz) console panel lighting bus.

3.7.14 ICS switch. The HOT MIC/VOX ON/NORM/ICS OFF switch shall be a rotary 4 position switch.

3.7.15 Monitor switches. The integral monitor switch/level control shall be a combination push-pull (on-off) switch and rotary potentiometer. The switch shall be configured such that depressing the switch will activate that monitor line and pulling out on the switch will inactivate the monitor line. Minimum volume will be achieved at full counter-clockwise rotation of the potentiometer and maximum volume at full clockwise rotation.

3.7.16 Interface wiring. The assignment of interface connector pins shall be as specified in Table I. All radio and interphone audio inputs and outputs of the unit are designed for balanced operation with respect to signal ground, using twisted, shielded pair wiring. The unit shall be designed so that a single transmit-receive set unbalance occurring in the system wiring (audio lows shorted to ground) will not deleteriously impact on the operational and isolation integrity of the unit.

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TABLE I. Pin connections.

J1 D6001897-5 Mates with MS 3126E20-41S	
Connector Pin	Function
C	28 VDC Power
D	Power Ground
R	Panel Light +5V
U	Panel Light Power Ground
e	Chassis Ground
H	Chassis Ground
L	Microphone Input (HI)
M	Microphone Input (LO)
K	Wideband Audio Out
b	Interphone Key
c	Key Common
d	Radio Key
f	Transmitter Control #1
g	Transmitter Control #2
h	Transmitter Control #3
r	Transmitter Control #4
t	Transmitter Control #5
X	Logic Select (PVT)
Y	Logic Select (ICS)
Z	Logic Select (XMIT 1)
i	Logic Select (XMIT 2)
j	Logic Select (XMIT 3)
m	Logic Select (XMIT 4)
n	Logic Select (XMIT 5)
N	Transmit Audio Out 1A
P	Transmit Audio Out 1B
S	Transmit Audio Out 2A
T	Transmit Audio Out 2B
V	Transmit Audio Out 3A
W	Transmit Audio Out 3B
k	Transmit Audio Out 4A
s	Transmit Audio Out 4B
p	Transmit Audio Out 5A
q	Transmit Audio Out 5B

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TABLE I. Pin connections - Continued.

J1 D6001897-5 Mates with MS 3126E20-41S	
Connector Pin	Function
A	PVT Hot Mic
B	Remote Xmit Sel 5
E	Panel Light +28V
F	Remote Xmit Sel 4
G	Aux Mic In
J	Panel Light 115V AC
a	Remote Xmit Sel 3

J2 D6001897-6 Mates with MS 3126E20-41SW	
Connector Pin	Function
C	Chassis ground
D	Headset audio (HI)
E	Headset audio (LO)
g	Private line PVT-A
P	Private Line PVT-B
R	Interphone Line ICS-A
S	Interphone Line ICS-B
A	Receiver Audio In 1A
B	Receiver Audio In 1B
V	Receiver Audio In 2A
W	Receiver Audio In 2B
U	Receiver Audio In 3A
i	Receiver Audio In 3B
X	Receiver Audio In 4A
Y	Receiver Audio In 4B
k	Receiver Audio In 5A
m	Receiver Audio In 5B
F	Navigation Receiver Audio 1A
G	Navigation Receiver Audio 1B
a	Navigation Receiver Audio 2A
b	Navigation Receiver Audio 2B
n	Navigation Receiver Audio 3A
p	Navigation Receiver Audio 3B
c	Navigation Receiver Audio 4A
H	Navigation Receiver Audio 4B
J	Navigation Receiver Audio 5A
K	Navigation Receiver Audio 5B
s	Navigation Receiver Audio 6A
t	Navigation Receiver Audio 6B

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TABLE I. Pin connections - Continued.

J2 D6001897-6 Mates with MS 3126E20-41SW	
Connector Pin	Function
N	Direct Audio (Uncontrolled) 1A
f	Direct Audio (Uncontrolled) 1B
q	Direct Audio (Uncontrolled) 2A
r	Direct Audio (Uncontrolled) 2B
L	Direct Audio (Controlled) 3A
M	Direct Audio (Controlled) 3B
d	Direct Audio (Controlled) 4A
e	Direct Audio (Controlled) 4B
Z	Remote Xmit Sel 1
T	Remote Xmit Sel 2
h	SEQUENCE/SP5T SEL
i	MASTER/SLAVE Control

3.8 Environmental service conditions. Each unit shall operate under the following environmental conditions without damage or loss of performance:

3.8.1 High temperature. The equipment shall operate at ambient air temperatures up to +145°F and withstand exposure (non-operating) to ambient air temperatures to +160°F.

3.8.2 Low temperature. The equipment shall operate at ambient air temperatures down to -25°F and withstand exposure (non-operating) to ambient air temperatures to -70°F.

3.8.3 Altitude. The equipment shall withstand air transportation in unpressurized aircraft up to 50,000 feet above sea level. The input supply voltages shall be from 17.0 to 33.0 volts dc.

3.8.4 Humidity. The equipment shall operate during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.5 Sand and dust. The equipment shall withstand, in both operating and nonoperating conditions, exposure to sand and dust particles that may accumulate within the equipment shelter.

3.8.6 Salt fog. The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.7 Fungus. The equipment shall operate during and after prolonged exposure to tropical climate fungus, rot, or mildew.

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3.8.8 Explosive atmosphere. When operating in an explosive atmosphere environment the unit shall not inflame a gaseous air mixture.

3.8.9 Rain. During periods of heavy precipitation with intermittent wind.

3.8.10 Acceleration. When exposed to steady-state acceleration levels of up to 6g for a duration of 1 minute.

3.8.11 Vibration. When subjected to helicopter vibration characterized by dominant sinusoidal superimposed on a broadband random background.

3.8.12 Shock. The equipment shall withstand shocks encountered during in-service use and field servicing.

3.8.13 Crash safety. The equipment shall withstand forces induced by in-service crash landings. A dummy mass with outside structure and mounting points identical to the flight hardware and mass-loaded to be dynamically the same as the flight hardware may be used for the crash safety test.

3.9 Voltage protection, low and reverse. The unit shall be subjected to input power requirements of paragraph 3.10 and incur no damage after being subjected to 33 volts dc reverse polarity and continuous operation at 17 to 33 volts dc.

3.10 Input power. The unit shall operate from normal 27.5 volt dc aircraft power and shall maintain required performance over the input voltage range of 22 to 33 volts dc. It shall also maintain required performance when subjected to input overvoltage. During emergency mode, 17.0 volts to 22.0 volts, and undervoltage conditions, the unit will operate with degraded performance. When the input current is measured with 29.0 volt dc input, 1.25 volts audio output at the headset pins (before transmit attenuation (see 3.29) and 0 dBm output on one transmitter audio output line, the CSC input current shall not exceed 250 milliamps, 7.25 watts. The input power requirements for front panel display lamps shall be no greater than:

a. 28 volts dc, 4.0 watts (red), 6.5 watts (green)

b. 5 volt ac/dc, 4.0 watts (red), 6.5 watts (green)

c. 115V ac, 400 Hz. 3 watts

3.11 Output interphone.

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3.11.1 ICS line. With the selector switch set in any position PVT, ICS, or 1 through 5, a 1.0 KHz signal at 0.6 volts applied to the required pad (see 4.3.7.11f), which is connected to pins J1-L and J1-M (microphone input) and pin J1-b jumpered to ground, the output across pins J2-R and J2-S shall be between .35 and .43 volts; this output shall also be the same for any remote channel selection and also when the HOT MIC-VOX ON-NORM-OFF switch is set to HOT-MIC or VOX ON with VOX control fully counterclockwise and pin J1-b is ungrounded. HOT MIC and Interphone key (pin J1-b) operation shall be overridden when any radio channel is selected and keyed. HOT MIC and VOX ON shall be overridden when PVT is selected and J1-A or J1-d is jumpered to ground.

3.11.2 PVT line. With the selector switch set in PVT position, a 1.0 KHz signal at 0.6 volts applied to the required pad (see 4.3.7.11f), to pins J1-L and J1-M (microphone input), the output across pins J2-q and J2-P shall be between .35 and .43 volts when pin J1-A is Jumpered to ground (J1-c). Pin J1-A may be grounded in the aircraft wiring to provide automatic HOT MIC activation when selector switch is set to PVT.

3.11.3 ICS off. Any ICS signal to the headset shall be reduced at least 60 dB when switch is set to ICS OFF position.

3.12 Sidetone. With selector switch set to ICS position, HOT MIC-VOX ON-NORM-ICS OFF switch to NORM position, pin J1-d or J1-b jumpered to ground and 1.0 KHz signal at 0.6 Volts applied to the required pad, the output voltage across pins J2-D and J2-E shall be -4.0 to -6.0 dB below the level obtained across J2-D and J2-E when a 0.39 volt rms, 1.0 KHz signal is applied to pins J2-R and J2-S. A comparable sidetone reduction shall occur when selector switch is set to PVT and pin J1-A is jumpered to ground.

3.13 Response, microphone amp. The frequency response shall not vary more than ±1.0 dB within the frequency range of 700 Hz to 2250 Hz. The 3 dB ±1.0 dB bandwidth shall be 350 Hz to 4500 Hz. The 1 dB and 3 dB ± 1 dB levels are referenced to the level at 1 kHz. The amplitude between 350 Hz and 4500 Hz shall be no greater than 1 dB above the amplitude at 1 KHz. The slope of the asymptotic attenuation curve above 4500 HZ and below 350 Hz shall be no less than 6 dB per octave.

3.13.1 Response, wideband audio out. The frequency response shall not vary more than ±1.0 dB in the range 200 to 3000 Hz. The 3 dB ± 1 dB bandwidth shall be 100 to 6000 Hz. These levels are referenced to the 1 KHz level. The slope of the asymptotic attenuation curve above 6000 and below 100 should be no less than 6 dB per octave.

3.14 Distortion, microphone amp. The total harmonic distortion shall not exceed three (3) percent with a signal level of 0.6 volts applied to the required pad and five (5) percent with a

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signal level of 4.0 volts applied to the required pad over the frequency range of 350 to 4500 Hz. Center clipping ("zerocrossing distortion") shall be no greater than 2 dB with a signal level of 0.085 volts at 1.0 KHz applied to the required pad.

3.15 Attack and release time, AGC. With 1.0 KHz applied to pins J1-L and J1-M, the attack time shall not be more than 0.1 seconds and the release time shall be 3.0 seconds ±1.5 seconds.

3.15.1 Attack and release time VOX. With 1.0 KHz applied to pins J1-L and J1-M, the attack time shall not be more than 0.1 seconds and the release time shall be between 0.5 and 1.5 seconds.

3.16 Breakpoint and slope, AGC. With 1.0 KHz applied to the pad connected to pins J1-L and J1-M (see 4.3.7.11f) and when the linearity of the input-output curve varies 1.0 dB (this will be the breakpoint), the input signal level shall be between 0.2 and 0.5 volts. With an increase in the input signal level from 0.5 to 5.0 volts the slope of the input-output curve from the breakpoint shall be 2 dB maximum change in output for a 20 dB change in input. This requirement applies to the equipment operated in both the 5 ohm and 150 ohm microphone input positions.

3.16.1 Sensitivity of VOX. The sensitivity of the VOX circuit shall be at least 8 dB greater at 500 Hz than at 3 KHz. At the maximum counterclockwise position of the VOX control and with the ICS switch at VOX ON the unit will operate exactly as it does for the ICS switch at HOT MIC.

3.17 Transmit audio output, microphone amp. With J2-h open and J1-d connected to J1-c (key common) or with J2-h connected to key common (manual SP5T mode) and any of-J2-Z, T or J1-a, F, B, d connected to key common, the transmit audio shall be enabled as follows: with the microphone selector switch in position 1 and the channel selector switch in positions 1 through 5 and a 1.0 KHz signal at a level of 0.6 volts applied to the 5000 to 5 ohm pad connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%. With the microphone selector switch in position 2 and a 1.0 KHz signal at a level of 0.6 volts applied to a 2.4K to 150 ohm pad with blocking capacitor, 47 pfd or greater, connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%. The wideband audio output J1-K shall be 28 mV ±10% for either position of the MIC switch.

3.17.1 Aux mic input. With an Input of 28mV at 1 KHz the output on the indicated output lines shall be 0.39V +l0%.

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3.17.2 Remote transmit selection. With the transmit select control operated to RMT the operation shall be as follows:

3.17.2.1 Sequential step mode. With power on and +28 volts supplied through a resistor to limit the current to less than 1 Ampere to each of pins Jl-f, q, h, r, t, transmit selection shall be accomplished sequentially, one channel at a time, by momentarily connecting pin J1-B to J1-c. Transmit selection shall be in the sequence 1, 2, 3, 4, 5, 1, 2, etc. Performance of the unit shall meet the same requirements as for manual selection of any channel. With +28 volts removed from J1-q, channel 2 shall be skipped and so on. If all above pins are disconnected from +28 volts, channel 1 shall be selected. At power turn-on the first channel with +28 volts on it will be selected.

3.17.2.2 SP5T switch mode. With pin J2-h connected to J1-c (key common), the unit shall be in receive mode unless one of the following pin connections is made:

With pin J2-Z connected to J1-c, channel 1 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 1 with J1-d connected to J1-c (Radio PTT ON).

With pin J2-T connected to J1-c, channel 2 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 2 with J1-d connected to J1-c (Radio PTT ON).

With pin J1-a connected to J1-c , channel 3 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 3 with J1-d connected to J1-c (Radio PTT ON).

With pin J1-F connected to J1-c, channel 4 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 4 with J1-d connected to J1-c (Radio PTT ON).

With pin J1-B connected to J1-c channel 5 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 5 with J1-d connected to J1-c (Radio PTT ON).

3.17.3 Remote indicator. The unit shall operate from 27.5 volt dc aircraft power and maintain performance over the entire range of 17.0 to 33.0 volts dc. The input power shall not exceed 2 watts at 28 volts dc and 2.5 watts at 5 volts ac/dc.

3.17.3.1 Weight. The weight of the completely assembled remote indicator unit shall not exceed 1.7 pounds.

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3.17.3.2 Outline dimensions. The outline dimensions of the completely assembled unit shall be within the following limitations and exclusive of the cable:

Panel/Housing Height - 1.875 inches maximum.

Panel width - 5.75 inches maximum.

Housing width - 5.0 inches maximum.

Housing depth (from mounting surface and less connector) - 2.9 inches maximum.

3.17.3.3 Chromaticity. Normal legend illumination shall be ANVIS GREEN A with a U' coordinate of 0.099, V' coordinate of 0.544 and a radius of 0.034 on the 1976 UCS chromaticity diagram measured at 0.1 foot-lambert display luminance.

3.17.3.4 Spectral radiance. The Anvis Radiance shall not exceed 1.7×10^{-10} AR when illuminated to produce 0.1 foot-lambert display luminance.

3.17.3.5 Chromaticity and spectral radiance. The chromaticity shall fall within the area noted as aviation green with a dominant wavelength of 510 ± 15 nanometers (nm) and allow no more than one percent of the relative peak amplitude to exist between 600 and 930 nm. Measurements to be made at 15.0 foot-lambert display luminance.

3.17.3.6 Input power. The unit shall be operated from normal 27.5 volt dc aircraft power and maintain required performance other than light intensity, over the entire range of 17 to 33 volts dc and when subjected to overvoltage. In normal operation (one annuniciator on) the input power shall be 28 volts dc (2 watts max) and 5 volts ac/dc (2.5 watts max). During lamp test the input power at 28 volts dc shall be 6 watts maximum.

3.17.3.7 Annuniciator control. when the appropriate control line is grounded the corresponding dual lamp anniciator shall light. When a voltage in the range of 17 to 33 volts is applied to the lamp test line all the annunciator lamps will light simultaneously but not at normal full brightness. When a ground is applied to the bright/dim line, the annunciator lamps which are on, will be bright.

3.17.3.8 Transient protection. The units shall meet specified requirements after subjection to a + 80 volt pulse for 0.2 second duration three times within one minute. The unit shall also meet specified requirements after subjection to a $\pm$ 600 volt spike of 5 μs $\pm$ 0.2 μs duration, applied three times within one minute.

3.17.4 Piggy back operation. Two units may be operated in tandem to provide capability for 10 radios, 12 NAV channels, four DIRECT AUDIO (two uncontrolled and two controlled) channels and two PVT channels and two ICS channels. Two units shall be configured for piggy back operations by external connections only. One of the two units will be selected for master

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and one for slave status. Master status shall be selected by externally connecting J1-a to J1c (key common). Slave Status shall be selected by connecting J2-T to J1-c and leaving J1-a open. Remote operation in a piggyback configuration shall be by sequential step mode only. When two units are configured for piggyback operation it shall be possible to operate with both units in manual, both units in remote, or one unit in manual and the other in remote.

3.17.4.1 Interphone operation. When the ICS is keyed by ICS PTT (J1-b to J1-c) or master unit HOT MICA or by master unit VOX or by RAD PTT with ICS selected on master unit, the master unit ICS audio circuit shall be selected and keyed unless the master unit is in remote and the slave unit is in manual with ICS selected.

3.17.4.2 Master unit in manual mode. When the master unit is in manual mode, the two unit Station shall meet all receive mode requirements for all receive channels. Channel selection shall be by the select switch of the master unit only.

3.17.4.3 Slave unit in manual mode. When the slave unit is in manual mode and the master unit is in manual mode, operation shall be in accordance with 3.13.8.3.2. When the slave unit is in manual mode and the master unit in remote mode, channel selection will be only by means of the slave unit select control. When a transmit channel is not being keyed the station will meet all the receive mode requirements for all receive channels. When either PVT or ICS is selected on the slave unit mic audio will be applied to the selected interphone line corresponding to the slave unit. When the PVT channel or a transmit channel is selected but not keyed and interphone is keyed mic audio will be applied to the ICS line corresponding to the master unit.

3.17.4.4 Both units in remote mode. When RMT is selected on both units transmit selection will be by sequential selection only (momentary ground to J1-B). Channel selection will be per paragraph 3.17.2.1, the next channel after channel 5 on any unit will be channel 1 on the other unit.

3.18 Audio output headset amp. The headset amp shall contain internally adjustable control to permit adjustment of the output level of the headset amp. With an input of 2.75 V, 1 kHz at any one radio, NAV and DIRECT channel or an input of 0.39V, 1 kHz at either the ICS or PVT interphone input, the output across 8 ohms shall be adjustable from below 1.0 volts to above 1.6 volts. With the MIC switch in position 2 (150 ohms) after the output level was previously set to 1.25 volts with MIC switch in position 1, the output level shall be 1.25V ±10%.

With the control set for 1.25 volts output across 8 ohms the amplifier output voltage shall be limited when the input voltage to a radio NAV or DIRECT channel cross a threshold between 2.75 and 3.87 volts. The limiter circuitry shall operate effectively through 5.47 volts of input, to limit the output to within 4 dB of the maximum set value of 1.25V. The output level of a

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received tone shall not be reduced by more than 9 dB by the presence of another tone greater than the limiting level at any RADIO, NAV, or DIRECT channel.

3.19 Response, headset amp. The frequency response shall not vary more than ±1.0 dB within the frequency range of 700 Hz to 2250 Hz. The 3 dB ± 1 dB bandwidth shall be 350-Hz to 4500 Hz. The 1 dB and 3 dB levels are referenced to the level at 1 KHz. The amplitude between 350 Hz and 4500 Hz shall not be greater than 1 dB above the amplitude at 1 kHz. The slope of the asymptotic attenuation curve above 4500 Hz and below 350 Hz shall be no less than 6 dB per Octave.

3.20 Distortion, headset amp. When frequencies over the range of 350 to 4500 Hz are applied with the required input voltages to audio input lines, the total harmonic distortion at the headset amp output shall not exceed ten (10) percent. Center clipping ("zero-crossing distortion") shall be no greater than 2 dB below the peak output level with a signal level of 0.39 volts at 1.0 KHz applied to the receive audio input lines or .055 volts applied to the interphone lines.

3.21 Broadband noise, headset. The headset broadband noise output of the unit(s) shall not exceed 0.9 millivolts when subjected to the test of 4.3.21.

3.22 Isolation, monitor switches/audio lines. With 1.0 KHz applied to the specified audio inputs and a 1.25V output at the headset amplifier, the output shall not exceed 1.25 mV when the monitor switch is set to the "OFF" position and the selector switch is set to any other audio input line. With 2.75 volts, 1.0 kHz applied to any audio input line, monitor switches to "ON" position, the audio level on any other audio line shall be less than 2.75 millivolts. (When the input signal is applied to one audio input, the signal shall be less than 27.5 millivolts on the two common NAV audio lines or the common DIRECT line.)

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3.23 Isolation, transmitter lines. With 2.75 volts 1.0 kHz applied to any monitor input line or 0.39 volts, 1.0 kHz applied to the ICS or PVT lines and the headset amplifier output set for 1.25V, the undesired voltage level at each transmitter audio output or control line shall be at least 120 dB below the normal 0.39 volts output, except that on the selected transmitter audio output line the undesired signal shall not be detectable in the output noise floor as observed using a 5 Hz bandwidth. The output noise floor as observed using a 5 Hz bandwidth shall be no greater than -90 dB referenced to 0.39V out. Similarly, a 1.0 kHz, 0.39 volt, signal on any selected ICS, PVT or transmit audio line shall have an isolation of at least 120 dB below normal, 0.39 volts output at any unselected transmit audio line.

3.24 Transient protection. The unit(s) shall meet specified requirements after subjection to a +80 volt pulse and d -80 volt pulse for 0.2 second duration three (3) times within 1 minute. In addition, the unit shall meet specified requirements after subjection to a +600-volt spike of 5-microseconds ±0.2 microseconds duration, applied three (3) times within 1 minute.

3.25 Headset amp, unterminated. The headset amp output shall be stable (no oscillation) when pins J2-D and J2-E are open circuited, or short circuited and grounded, and the monitor and selector switches are placed to any position and all volume controls are rotated to any level between minimum and maximum audio output.

3.26 Transmitter control. With any of radio channels 1 through 5 selected, pin J1-d jumpered to ground, and a dc voltage applied to the specified pins, the transmitter control lines shall withstand 1.0 amps through the lines without any degradation.

3.27 Function selector logic. With the selection of any one of PVT, ICS, 1,2,3,4, or 5, a separate .5-ampere capacity circuit pin to ground shall be provided. The normally open pin shall be grounded (dc Power Gnd) when its respective function is selected.

3.28 Microphone power. The unit shall provide up to 10 mA at 10-volts dc ±20% on the microphone high input (J1-L) with power return through the microphone low input (J1-M) when the microphone selection switch is set to position 2.

3.29 Received audio input attenuation during transmit. With a 0.39 volt rms, 1.0 kHz input applied to PVT or ICS input or a 2.75 volt rms, 1.0 kHz input applied to any other audio input and the headset amplifier output set for 1.25V rms, the received audio inputs appearing at the headset amplifier output shall be selectively attenuated 12 dB ± 1 dB (below the 1.25Vrms level) as-shown in TABLE II when selector switch is set to positions 1 through 5 and pin J1-d is grounded. With J1-F and J1-d both grounded to key common, all the received audio inputs except DIRECT channels shall be attenuated 12 dB ± 1 dB.

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3.30 Audio level (volume) adjustments.

3.30.1 Main volume control. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 40 ± 3 dB when the main volume control is set for minimum audio output.

3.30.2 Monitor audio level adjustment. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 27 ± 2dB when the individual audio level control (for the selected radio audio monitor input) is set for minimum audio output. The audio outputs of all other monitor lines shall be unaffected by this adjustment. This requirement shall be met for any headset amplifier adjustment level between 1.1 VRMS and 1.6 VRMS.

TABLE II. Received audio input attenuation.

Selector Switch Position	PVT	ICS	Received Audio Level					Nav 1	Nav 2	Direct radios
			1	2	3	4	5			
1	-12 dB	-12 dB	0 dB	-12 dB					-12 dB	0 dB
2	-12 dB		-12 dB	0 dB	-12 dB				-12 dB	0 dB
3	-12 dB			-12 dB	0 dB	-12 dB			-12 dB	0 dB
4	-12 dB				-12 dB	0 dB	-12 dB		-12 dB	0 dB
5	-12 dB					-12 dB	0 dB	-12 dB	-12 dB	0 dB
PVT	Sidetone	0 dB								0 dB
ICS	0 dB	Sidetone	0 dB							0 dB

3.31 TEMPEST. The equipment shall comply with the requirements of NACSEM 5100 titled "TEMPEST Requirements" for the Control Unit, Communication System C-11746(V)( )/ ARC.

3.32 Electromagnetic interference. The performance of the system shall not be degraded nor interfere with other equipment while operating in a military electromagnetic environment or be adversely affected by existing electromagnetic interference.

3.32.1 Audio susceptibility. The performance of the equipment shall not be degraded when subjected to electromagnetic energy injected on its dc power leads

3.32.2 Conducted susceptibility. The performance of the equipment shall not be degraded in the frequency range of 50 KHz to 400 MHz when subjected to the specified power levels except that all RF signals be modulated 50 percent AM at 1.0 KHz.

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3.32.3 Radiated susceptibility. The performance of the equipment shall not be degraded in the applicable frequency range except that all RF signals shall be modulated 50 percent AM at 1.0 kHz:

3.32.4 Radiated susceptibility induction field. The unit headset noise output (J2-D and J2-E) shall not exceed 10 mV and the interphone noise output (J2-R and J2-S) shall not exceed 20 mV.

3.33 Bonds, grounds, connectors and cables. Bonding for the equipment current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case and chassis mounted connectors shall have approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Interconnecting cables requiring shields shall have a minimum of 40 dB shield attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. The shield shall be constructed of tinned copper braid. Connectors with cables having shields shall have an approved RF conductive finish and provided with EMI backshells for terminating the shields.

3.34 Operational inspection. Each unit shall be operated as specified after applicable verification tests are completed to determine that each unit is operable prior to packaging. Voice quality shall indicate proper operation of the unit.

3.35 Interchangeability/interoperability. Like units, assemblies, subassemblies, modules, and replaceable parts built to this performance specification shall be physically and functionally interchangeable without modification and degradation of performance. These same units shall also be physically and functionally interchangeable with like units of the “standard” Communication System Control Unit C-11746 units without modification and degradation of performance.

4. VERIFICATION

4.1 Classification of inspections. Inspection requirements are classified as follows:

a. First article inspection (see 4.3.1).

b. Conformance inspection (see 4.3.2).

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4.2 Verification methods. The types of verification methods in this section can be visual inspection, measurement, sample tests, demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, materials analysis, and similarity to previously approved or previously qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.3 Order of inspection. Perform 100% visual mechanical and electrical inspection of assemblies and subassemblies first, prior to their integration into the end item. Upon successful completion of this test, perform 100% visual mechanical and electrical inspection of the end items. Upon successful completion of this test, perform conformance test as specified in the contract or purchase order. Upon successful completion of conformance test, perform all verification tests in Table III.

4.3.1 First article inspection. First article inspection shall be performed on that number of complete units as specified in the contract or purchase order. When a first article sample is required (see 3.1 and 6.3), the units shall be produced with equipment and procedures used in production (production-representative units). This inspection shall include the verification tests listed in Table III. Failure of any inspection or test shall be cause for rejection of first article inspection.

4.3.2 Conformance inspection. Conformance inspection shall include those examinations and tests from Table III as defined in the contract (see 6.4).

4.3.3 Finish inspection. Visually inspect and verify external colors of all materials to be a dark color (i.e., black, olive drab, brown, etc.) to facilitate existing military camouflage schemes. Make ten measurements at different points on the system surface using a 60 degree glossmeter. Confirm the average value of these ten readings does not exceed ten percent.

4.3.4 Marking inspection. Inspect and verify the presence of the system designator, contract number, unique serial numbers on each nomenclatured item and CCA, date, manufacturer code, part number, and national stock number in clear and legible lettering. Verify that the method used to attach labeling to assemblies, subassemblies, and end items show no signs of breakage resulting from exposure to environmental testing.

4.3.5 Electrical preconditioning. Subject the equipment to a 48 hour on-time electrical preconditioning test. Run the unit continuously for 48 hours, the last 24 hours shall be failure free. Turn power OFF for 2 hours. Turn equipment ON and perform operational checkout of equipment.

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4.3.6 Safety inspection. Visually inspect assemblies, subassemblies, and end items for the presence of protrusions, points, sharp edges or parts, chips, nicks, burrs, foreign material, harmful or extraneous materials, smudges, loose spattered or excess solder posing physical danger to personnel or equipment. Inspect for evidence of contact of wire or cable, broken strands, splicing, chafing, fraying, breaks, bends, cracks, strain, discontinuity of shielding of jacket material, blistering, exposed conductors, suitable clearances, loose or open components, or other indications of damage posing potential electrical danger to personnel or equipment. Inspect for other flaws detracting from the physical uniformity and neat appearance of the equipment.

4.3.7 Operating requirements verification.

4.3.7.1 Remote control. Demonstrate the equipment is be capable of accommodating two types of remote control. In the sequential mode, transmit selection shall be by means of an external momentary contact switch. In the SP5T mode, transmit selection shall be accomplished by actuating one of five momentary contact switches, one for each radio channel. Operation of one of the switches shall also key the selected radio. An SP5T switch has five positions, namely; up, down, left, right, center-in.

4.3.7.2 Piggy back configuration. Demonstrate the equipment is capable of being configured in tandem with another unit to provide selection and control of 10 radios at a single station. This configuration shall be accomplished by external connections only. Remote sequential mode control shall be operable in this configuration.

4.3.7.3 VOX. Demonstrate the equipment is capable of voice actuated mic keying for the ICS channel only. This feature will be selectable from the front panel HOT MIC/VOX ON/NORM/ICS OFF switch. In addition, VOX threshold control shall be available on the front panel.

4.3.7.4 Weight. Weigh a completely assembled unit to determine compliance.

4.3.7.5 Outline dimensions. Measure and verify the dimensions of the completely assembled unit conform to the requirements of Military Standard MS 25212 (ASG) within the following limitations:

Panel/Housing Height - 2.625 inches (6.67 cm)

NOTE: Housing height behind front panel such that dimension .048 min. between top of panel and top of housing and between bottom of panel and bottom of housing.

Panel Width - 5.75 inches (14.60 cm)
Housing Width - 5.0 inches max. (12.77 cm)
Housing Depth - 5.12 inches (11.75 cm) (excluding front panel and rear connectors)

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4.3.7.6 Securing of parts. Demonstrate Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified environmental conditions.

TABLE III. Verification methods.

Title	Requirements	Verification
Finish	3.3	4.3.3
Marking	3.4	4.3.4
Electrical preconditioning	3.5	4.3.5
Safety	3.6	4.3.6
Operating requirements	3.7	4.3.7
Remote control	3.7.1	4.3.7.1
Piggy back configuration	3.7.2	4.3.7.2
VOX	3.7.3	4.3.7.3
Weight	3.7.4	4.3.7.4
Outline dimensions	3.7.5	4.3.7.5
Securing of parts	3.7.6	4.3.7.6
Front panel	3.7.7	4.3.7.7
Reliability	3.7.8	4.3.7.8
Chromaticity and spectrum	3.7.9	4.3.7.9
Brightness	3.7.10	4.3.7.10
Power	3.7.11	4.3.7.11
Panel protection	3.7.12	4.3.7.12
Alternate front panel red	3.7.13	4.3.7.13
ICS switch	3.7.14	4.3.7.14
Monitor switch	3.7.15	4.3.7.15
Interface wiring	3.7.16	4.3.7.16
Environmental service conditions	3.8	4.3.8
High temperature	3.8.1	4.3.8.1
Low temperature	3.8.2	4.3.8.2
Altitude	3.8.3	4.3.8.3
Humidity	3.8.4	4.3.8.4
Sand and dust	3.8.5	4.3.8.5
Salt fog	3.8.6	4.3.8.6
Fungus	3.8.7	4.3.8.7
Explosive atmosphere	3.8.8	4.3.8.8
Rain	3.8.9	4.3.8.9
Acceleration	3.8.10	4.3.8.10
Vibration	3.8.11	4.3.8.11
Shock	3.8.12	4.3.8.12
Crash safety	3.8.13	4.3.8.13
Voltage protection, low and reverse	3.9	4.3.9
Input power	3.10	4.3.10.1
Output interphone	3.11	4.3.11
ICS line	3.11.1	4.3.11.1

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TABLE III. Verification methods - Continued.

Title	Requirements	Verification
PVT line	3.11.2	4.3.11.2
ICS off	3.11.3	4.3.11.3
Sidetone	3.12	4.3.12
Response, microphone amp	3.13	4.3.13
Response wideband audio out	3.13.1	4.3.13.1
Distortion, microphone amp	3.14	4.3.14
Attack and release time, AGC	3.15	4.3.15
Attack and release time, VOX	3.15.1	4.3.15.1
Breakpoint and slope, AGC	3.16	4.3.16
Sensitivity of VOX	3.16.1	4.3.16.1
Transmit audio output, microphone amp	3.17	4.3.17
Aux mic input	3.17.1	4.3.17.1
Remote transmit selection	3.17.2	4.3.17.2
Sequential step mode	3.17.2.1	4.3.17.2.1
SP5T switch mode	3.17.2.2	4.3.17.2.2
Remote indicator	3.17.3	4.3.17.3
Weight	3.17.3.1	4.3.17.3.1
Outline dimensions	3.17.3.2	4.3.17.3.2
Chromaticity	3.17.3.3	4.3.17.3.3
Spectral radiance	3.17.3.4	4.3.17.3.4
Chromaticity and spectral radiance	3.17.3.5	4.3.17.3.5
Input power	3.17.3.6	4.3.17.3.6
Annunciator control	3.17.3.7	4.3.17.3.7
Transient protection	3.17.3.8	4.3.17.3.8
Piggy back operation	3.17.4	4.3.17.4
Interphone operation	3.17.4.1	4.3.17.4.1
Master unit in manual mode	3.17.4.2	4.3.17.4.2
Slave unit in manual mode	3.17.4.3	4.3.17.4.3
Both units in remote mode	3.17.4.4	4.3.17.4.4
Audio output, headset amp	3.18	4.3.18
Response, headset amp	3.19	4.3.19
Distortion headset amp	3.20	4.3.20
Broadband noise, headset	3.21	4.3.21
Isolation, monitor switches/audio lines	3.22	4.3.22
Isolation, transmitter lines	3.23	4.3.23
Transient protection	3.24	4.3.24
Headset amp, unterminated	3.25	4.3.25
Transmitter control	3.26	4.3.26
Function selector logic	3.27	4.3.27
Microphone power	3.28	4.3.28
Received audio input attenuation (Xmit)	3.29	4.3.29
Audio level adjustments	3.30	4.3.30

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TABLE III. Verification methods - Continued.

Title	Requirements	Verification
Main volume control	3.30.1	4.3.30.1
Monitor audio level adjustment	3.30.2	4.3.30.2
TEMPEST	3.31	4.3.31
Electromagnetic interference	3.32	4.3.32
Audio susceptibility	3.32.1	4.3.32.1
Conducted susceptibility	3.32.2	4.3.32.2
Radiated susceptibility	3.32.3	4.3.32.3
Radiated susceptibility induction field	3.32.4	4.3.32.4
Bonds, grounds, connectors and cables	3.33	4.3.33
Operational inspection	3.34	4.3.34
Interchangeability	3.35	4.3.35

4.3.7.7 Front panel (green). Apply an adjustable 115 volt, 400 Hz ac power source applied to pins J1-J and J1-U, or an adjustable 28V dc applied to pins J1-E and J1-U or an adjustable 5V dc/ac (400 Hz) applied to J1-R and J1-U, as applicable, the unit shall meet the requirements for front panel (green).

4.3.7.7.1 Alternate front panel (red). With an adjustable 28 volts dc or an adjustable 5 volts dc/ac (400 Hz) power source applied to pins J1-E and J1-U or pins J1-R and J1-U, as applicable, the unit(s) shall meet the requirements for alternate front panel (red).

4.3.7.8 Reliability testing. The system reliability shall he demonstrated the MTBF of 3500 hours on two samples under the following conditions: ambient temperature ranging from -54°C to +55°C, duty cycle of 100 percent (5 minutes receive, 1 minute transmit), and an airborne environment. The reliability qualification test conditions shall for combined environments for Turboprop Aircraft and Helicopter equipment.

4.3.7.9 Chromaticity and spectrum. Demonstrate normal legend illumination shall be ANVIS GREEN A with a U' coordinate of .099, V' coordinate of .544 and a radius of .034 on the 1976 UCS chromaticity diagram measured at 0.1 ft. lambert display luminance. The spectral radiance shall be such that the ANVIS RADIANCE (AR) shall not exceed 1.7×10^{-10} AR when illuminated to produce 0.1 ft. lambert (FL) display luminance.

4.3.7.10 Brightness. Demonstrate a brightness of 1.0 ± 0.5 Foot-Lamberts at nominal panel supply voltage.

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4.3.7.11 Power. Apply plus 17 and 33 volts dc to pins J1-C and J1-D for 5 minutes at each voltage, and then apply 33 volts dc with plus to pin J1-D and negative to pin J1-C for 1 minute. The unit(s) shall meet the requirement for voltage protection. Panel lighting shall be operable from the 28 volt dc or 5 volt dc/ac (400 Hz) aircraft source. Green panel lighting equipment shall also be operable from the 115V ac (400 Hz) or 28 volt dc aircraft source.

NOTE: Standard test conditions. The contractor shall maximize the tailoring of environmental test conditions to the individual equipment applications in addition to the following standard test conditions:

a. Input power: 27.5 ± 0.5-volts dc.

b. Operational duty cycle: One (1) minute talk, five (5) minute listen.

c. Warm-up time: Not more than 2 minutes.

d. Output loads: Headset amp; 8 ohms. Transmit audio output; 150 ohms (see 4.3.7.11g).

e. Input loads: Audio lines and interphone; 150 ohms (see 4.3.7.11g).

f. Input load microphone; MIC switch in position 1, 5,000 to 5 ohm pad; MIC switch in position 2, 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater (see 4.3.7.11g).

g. Connector (J1) Pins: Pin pairs N-P, S-T, V-W, k-s, p-q are each terminated with a 150-ohm resistor. When MIC switch is in position 1 pins L and M should be terminated with a 5K to 5-ohm resistor pad with the junction of the 5K and 5-ohm resistors connected to pin L. When MIC switch is in position 2. Pins L and M should be terminated with 2.4K and 150 ohm pad with blocking capacitor, 47 µfd or greater, with the junction of the 2.4K and 150-ohm resistors connected to Pin L through the blocking capacitor. Pins b, c, and d, are connected to a single pole, double-throw switch, with pin c connected to the center pole. All shields are connected to pin H (chassis ground). Pin J1-K terminated with 600 ohms to power ground.

h. Connector (J2) Pins: Pin pairs A-B, V-W, U-j, X-Y, k-m, s-t, F-G, a-b, n-p ., c-H, J-K, d-e, L-M, q-r, N-f, R-S, q-P are each terminated with a 150-ohm. resistor. Pin pair D and E are terminated with an 8-ohm resistor. Shields should be tied to pin C.

4.3.7.12 Panel protection. Verify the CSC design for each each panel type, i.e., 5 volt, 28 volt, or 115 volt, to ensure power is received over a separate power input pin, both at the input connector J1 and, at the panel connector. Verify the CSC can not be inadvertently installed in the incorrect aircraft.

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4.3.7.13 Alternate front panel (red). With an adjustable 28 volts dc or an adjustable 5 volts dc/ac (400 Hz) power source applied to pins J1-E and J1-U or pins J1-R and J1-U, as applicable, the unit(s) shall meet the requirements for alternate front panel (red).

4.3.7.14 ICS switch. Verify any ICS signal to the headset is reduced at least 60 dB when switch is set to ICS OFF position.

4.3.7.15 Monitor switches. Verify switch to be a combination push-pull (on-off) switch and rotary potentiometer. Demonstrate that depressing the switch will activate the monitor line and pulling out on the switch will inactivate the monitor line. Verify minimum volume will be achieved at full counter-clockwise rotation of the potentiometer and maximum volume at full clockwise rotation.

4.3.7.16 Interface wiring. Verify assignment of interface connector pins to be as specified in TABLE 1. Verify all radio and interphone audio inputs and outputs of the unit are designed for balanced operation with respect to signal ground, using twisted, shielded pair wiring. Verify that a single transmit-receive set unbalance occurring in the system wiring (audio lows shorted to ground) will not deleteriously impact on the operational and isolation integrity of the unit.

4.3.8 Environmental test conditions. Electrical tests performed before, during and after environmental testing shall include those tests required to determine compliance with output interphone, response microphone amp, distortion microphone amp, transmit audio output microphone amp, audio output headset amp, response headset amp, distortion headset amp, headset amp unterminated, transmitter control, and function selector logic. Each unit shall operate under the following environmental test conditions without damage or loss of performance.

4.3.8.1 High temperature. The equipment in its operational configuration, shall be subjected to high temperature cycling ranging from natural ambient conditions to +145°F and +160°F. Raise the temperature to +145°F within a time period of one hour. During the maximum response period of exposure, perform a complete operational checkout of the equipment. Maintain this condition for six hours. Upon completion, turn the equipment OFF and raise the temperature to +160°F within a time period of one hour and maintain at that temperature for four additional hours (non-operating). Lower the temperature to +145°F within a time period of one hour. Following stabilization of the equipment, turn the equipment ON and perform an operational checkout. Repeat this cycle two additional times for a total of three complete cycles.

4.3.8.2 Low temperature. The equipment in its operational configuration, shall be subjected to low temperature cycling ranging from natural ambient conditions to -25°F and -70°F.

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Lower the temperature to -25°F within a period of one hour. Operate the equipment until temperature stabilization is reached. Maintain this condition for six hours. Upon completion, turn the equipment OFF and lower the temperature to -70°F within a time period of one hour and maintain at that temperature for an additional four hours (non-operating). Raise the temperature to -25°F within a time period of one hour. Following temperature stabilization of the equipment, turn the equipment ON and perform a complete operational checkout. Repeat this cycle two additional times for a total of three complete cycles.

4.3.8.3 Altitude. The unit(s) shall be subjected a 1-hour exposure (non-operation) to a pressure of 3.4 inch of Hg, simulating air-transport at 50,000 feet. Adjust the air pressure to standard ambient atmospheric conditions over a period of 2 hours. Conduct a visual inspection for damage and perform a complete operational checkout of equipment.

4.3.8.4 Humidity. Subject the equipment to natural cyclic high humidity conditions under randomly varied temperatures between 79°F and 95°F with relative humidity varied between 97% and 100% with condensation. Maintain this condition for 6 hours while operating the equipment. Maintain 85% relative humidity and reduce the temperature to 86°F in eight hours. Maintain this condition for an additional eight hours while operating the equipment. Repeat these steps for a total of ten cycles. Near the end of the tenth cycle perform a complete operational check of the equipment. There shall be no mechanical deterioration as evidenced by corrosion of metal parts or degradation of performance.

4.3.8.5 Sand and dust. With the temperature at least 73°F with a relative humidity (maintained throughout the test) of no more than 30%, subject the sample to a blowing sand and dust concentration (97 to 99% by weight) of fine sand and dust particles ranging from 100 to 325 microns at air velocity of 200 +100 feet per minute for at least 3 hours. Stop the test, adjust the temperature to 140°F, and continue test article exposure for another 3 hours. The equipment shall be reoriented periodically to expose all its sides to the dust and sand stream. Stabilize at natural ambient conditions. Before the item is cleaned perform test of all switches. Clean the item of excess sand and dust and perform an operational checkout of the equipment.

4.3.8.6 Salt fog. Subject a test sample to a continuously atomized, finely divided, wet, dense salt spray mixture of 5% sodium chloride and 95% water. Uniformly distribute the salt fog over the test sample at a fallout rate between 0.3 and 3ml/80cm^2/hr for at least 96 hours at a constant 95°F with minimal air circulation. After 48-hours stop the test and perform a test of all switches. After another 48 hour the equipment shall show no signs of corrosion or any visible degradation.

4.3.8.7 Fungus. Satisfy the fungus requirement by one of the two following methods:

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a. Fungus resistance demonstration alternative. Perform the ASTM G 21 fungus resistance test.

b. Fungus resistance materials analysis. Verify by analysis or evaluation assemblies contain only fungus inert materials or treated to make them fungus inert or resistant.

4.3.8.8 Explosive atmosphere. Verify the equipment will not ignite a gaseous air mixture in an explosive environment when performing ON-OFF switch and electrical operations.

4.3.8.9 Rain. The equipment with cables in place shall be exposed to a "steady state" rainfall of 4 ± 1 inches per hour on all surfaces of the test items for a total time duration of 2 hours. After completion of the rain test, the units shall be turned-on within 10 minutes after test completion. The units shall be operated and tested to determine satisfactory system operation.

4.3.8.10 Acceleration. Demonstrate that the equipment will operate without degradation during and after being subjected to loads induced by in-service forward acceleration. The test level shall be 7g's in the up direction, 2 g's fore and aft direction, 3g's down, and 4 g's in each lateral direction. Maintain levels in each direction for at least 1 minute.

4.3.8.11 Vibration. The equipment when normally mounted (with vibration isolators in place, if any). Whenever possible measured data in terms of vibration levels should be collected and used based upon actual field use. The relative amplitudes of the sinusoids depend on the proximity of the equipment to the vibration excitation source test duration shall be four (4) hours of excitation for each of three mutually perpendicular axes for a total test time of 12 hours per component. If this is not feasible use the vibration levels in the range of .0010 to 1.0 power spectral density (g^2/Hz) over a frequency range of 5 Hz truncated at 500 Hz and test successively for four hours in three mutually perpendicular directions for a total of 12 hours. The total excursion of the applied vibration shall not be less than 0.030 inch. In each of three directions, change the frequency in steps of one Hertz and maintain each frequency for at least 10 seconds. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement. After testing, the equipment shall be opened and there shall be no evidence of deterioration of external and internal parts. Perform operational checkout of the equipment.

4.3.8.12 Shock. Subjected the equipment to 18 impact shocks of 15g consisting of three shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 millisecond. The "g" value shall be within ± 10% when measured with a 0.2 to 250 Hz filter, and maximum "g" value shall occur at approximately 5 ±1/2 milliseconds.

4.3.8.13 Crash safety. The equipment shall withstand 12 impact shocks of 30 g' s, consisting of two shocks in opposite directions along each of three mutually perpendicular axes.

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Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The “g” value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum “g” value shall occur at approximately 5 ± 1/2 milliseconds. Bending and distortion shall be permitted, however, there shall be no failure to the attaching joints and the equipment or dummy load shall remain in place

4.3.9 Conditions precedent to preconditioning and reliability testing. Prior to doing any preconditioning or reliability acceptance testing, the contractor shall perform a thermal and vibration design analysis backed by a thermal and vibration survey. The thermal survey shall be performed on one sample under the temperature and duty cycle specified in the reliability test. The thermal survey shall be used to identify hot spots and the component of greatest thermal inertia, and to establish the time-temperature relationship between the equipment and the test chamber air. These relationships shall be used to determine the level of equipment thermal stabilization. Should the results of the thermal survey indicate local temperatures significantly higher than those predicted by the final thermal design analysis or greater than those used for derating in the reliability prediction, corrective action shall be accomplished, verified, and approved prior to the start of reliability testing. A vibration survey shall also be performed on one sample of the equipment to search for resonant conditions and design weaknesses. Any failures which occur during vibration survey, corrective action shall be accomplished, verified, and approved prior to the start of reliability testing.

4.3.10 Electrical test.

4.3.10.1 Input power. With the equipment operating under standard test conditions except 29.0 Vdc shall be applied to Pins J1-C and J1-D. Adjust the input signal level at pins J2-R and J2-S for 1.25 volts across J2-D and J2-E. Jumper Pin J1-d to ground and apply 0.6 volts at 1.0 kHz to a 5000-ohm to 5-ohm pad connected to Pins J1-L and J1-M. While these inputs are applied, measure the dc current to determine compliance for input power. Reduce the input voltage to 21 Vdc at pins J1-C and J1-D. Verify that the performance requirements for output interphone, distortion microphone amp, transmit audio output microphone amp, audio output headset amp, distortion headset amp, and function selector logic are met. Raise the input voltage to 33 VdC at pins J1-C and J1-D. Verify that the performance requirements for output interphone, distortion microphone amp, transmit audio output microphone amp, audio output headset amp, distortion headset amp, and function selector logic are met. Connect 28 Vdc to pins J1-E and J1-U or 5V/400 Hz to J1-R and J1-U or 115V/400 Hz to J1-J and J1-U, as applicable, and measure the input current to determine compliance with input power.

4.3.11 Output, interphone.

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4.3.11.1 ICS. Set MIC switch in position 1 (5 ohms). Apply 0.6 Volts at 1.0 kHz to a 5,000 to 5 ohm pad connected to pin J1-L, M and set selector switch to ICS.

4.3.11.1.1 RADIO PTT. Jumper pin J1-d to ground (J1-c) and measure the output voltage at J2-R, S to verify ICS level per ICS line requirement.

4.3.11.1.2 ICS PTT. Switch jumper to J1-b to ground and measure output voltage for positions PVT, ICS and positions 1 through 5 of the selector switch. Set selector switch at RMT and momentarily jumper pin J1-B to ground to select Xmit channels 1 through 5 continuing to measure output voltage at J2-R, S to verify ICS level per ICS line requirement. Jumper pin J2-h to ground and measure output voltage at J2-R, S to verify ICS line requirement. Momentarily also ground J2-T, Z and J1-a, F, B one at a time and note output voltage drops to less than 10 mV. Remove jumper from J2-h to ground, and jumper J1-d to ground. Note that output voltage drops to less than 10 mV. Manually select channels 1 through 5 and note output also drops to less than 10 mV.

4.3.11.1.3 HOT MIC. Set HOT MIC-VOX ON-NORM-ICS OFF switch to HOT MIC and repeat tests of ICS PTT test.

4.3.11.1.4 VOX ON. Set HOT MIC-VOX ON-NORM-ICS OFF switch to VOX ON and set VOX control port fully counterclockwise. Repeat HOT MIC test.

4.3.11.2 PVT line. Apply .6 volts at 1.0 kHz to a 5,000 ohm to 5.0 ohm pad connected to pin J1-L and J1-M and set selector switch to PVT position and MIC switch to position 1 (5 ohms). Jumper pin J1-A to ground and measure output voltage at J2-g to verify compliance with PVT level. Repeat above for MIC and VOX ON positions of ICS switch. Return ICS switch to NORM. Remove jumper from J1-A and jumper J1-d to ground to again verify previous output voltage. Leave jumper J1-d connected to ground and also jumper J1-A to ground and rotate selector switch through positions ICS and 1 through 5 and note output voltage drops to less than 10 mV. Remove jumper from J1-d and rotate selector switch to RMT. Momentarily ground J1-B repetitively to select channels 1 through 5 again note output remains at less than 10 mV. Also Jumper J2-h to ground and momentarily ground J2-Z and T and J1-a, F and B one at a time noting that output remains at less than 10 mV.

4.3.11.3 ICS off. Apply a 1.0 kHz input signal at .39 volts rms to pins J2-R and S. Rotate selector knob to ICS and verify 1.25 volts at J2-D and E. Rotate ICS switch to ICS OFF and verify 1.25 Volts at J2-D and E. Rotate select knob to PVT and then through positions 1 through 5 and RMT. Verify that the voltage drops to less than 10 mV at all positions of the selector switch except ICS. These measurements shall be made to determine compliance.

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4.3.12 Sidetone. Apply 0.6 volts at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to ICS position and MIC switch to position 1 (5 ohms). Jumper pin J1-b to ground and measure the output voltage across pins J2-D and J2-E. Move jumper to pin J1-A to ground, set selector switch to PVT position and measure the output voltage at pins J2-D and J2-E. Remove the jumper and apply 0.39 volts reference signal to pins J2-R and J2-S and measure the output voltage. These output voltages shall be compared to determine compliance for sidetone.

4.3.13 Response, microphone amp. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position, jumper pin J1-d to ground, apply 0.1 volt at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and using the output level obtained across pins J2-R and J2-S as a reference level, measure the frequency response from 350 to 4,500 HZ and at 87 Hz and 18,000 Hz to determine compliance. Repeat above after setting MIC switch to position 2 (150 ohms) and using a 2.4K ohm to 150 ohm pad with blocking capacitor, 47 µfd or greater instead of a 5,000 ohm to 5 ohm pad.

4.3.13.1 Response, wideband audio output. Set-up unit as in the paragraph above for MIC switch position 1 except use output across pins J1-K (HI) and c (LO) as a reference level. Measure the frequency response from 100 to 6,000 Hz and at 40 and 15,000 Hz to determine compliance.

4.3.14 Distortion microphone amp.

4.3.14.1 Distortion microphone amp, HI voltage. Apply 4.0 volts at 350 Hz and 4.5 kHz to pins J1-L and J1-M through a 5000-ohm to 5-ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control to maximum audio output position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion to determine compliance.

4.3.14.2 Distortion microphone amp, additional. Apply 0.6 and then 4.0 volts at 1.0 kHz to pins J1-L and J1-M through a 5000 ohm to 5 ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control maximum CW and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion at each of the input voltages at 350 Hz, 1000 Hz and 4500 Hz to determine compliance. Reduce the input voltages to .085 volts at 1.0 kHz and measure the zero crossing distortion to determine compliance.

4.3.15 Attack and release time, AGC. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and apply 1.34 Volts

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at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S for a reference level. Remove the input signal for 30 seconds and reapply the input signal voltage, measure the AGC attack time required for the output voltage to reach within 1.0 dB of the reference level. Lower the input signal voltage to .85 ± 10%V, then reduce input voltage by 16 ± 1 dB and hold for 2 seconds. Measure output voltage for a reference level. Increase the input voltage by removing the 16 dB attenuation. Hold for 5 seconds and then insert the attenuation. Measure the release time required for the input voltage to reach within 1 dB of the reference voltage to determine compliance.

4.3.15.1 Attack and release times, VOX. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and ICS switch to VOX ON. Apply 1 volt from an audio generator to the 5,000 to 5 ohm pad at 1 kHz. Use one channel of a two channel oscilloscope to monitor the input to the pad. Use the other scope channel to monitor J2-R and S. Adjust the VOX control until the MIC amplifier is enabled but a slight CCW rotation of the knob would disable it. Instantaneously reduce the audio input level by a 10 dB step attenuator and note that when the input voltage is reduced 10 dB the MIC amplifier is disabled, and when increased back to 1v the MIC amplifier is enabled as seen by the output signal at J2-R, S. Set input at 1 volt and with a 1 cm/sec. sweep speed on the scope instantaneously decrease the input by 10 dB. The difference in time between the onset of the reduction in input level and the switch-off of the output signal is the VOX release time. Then set input at -10 dB reference to 1v and suddenly switch to 1 volt. The difference in time between the sudden increase in input level and the switch-on of the output signal is the VOX attack time.

4.3.16 Breakpoint and slope, AGC. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position and apply 0.1 volt at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S. Increase the input voltage to 0.2 volt then to 3.0 volts and measure the output voltages. Compare the input-output voltages to determine the breakpoint. Utilizing the output voltage obtained with 0.5 volts input as a reference level, increase the input signal level to 5.0 volts and measure the output voltage, compare the input-output voltages to determine the slope. Switch the microphone selector switch to the 150 ohm position and repeat above with 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater connected to pins J1-L and J1-M. The comparisons shall be made to determine compliance.

4.3.16.1 Sensitivity of VOX. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position. Apply 3 kHz at 1.1 volts to the 5,000 to 5 ohm pad. Set ICS switch to VOX

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ON and monitor ICS output (J2-R, S). Set VOX control fully CCW and observe ICS output on scope. Now rotate VOX control CW until ICS output just turns on. Adjust input to pad and reset VOX control if necessary until ICS output is obtained for 1.2 Volts and is shut off for 1.0 volts in. Change input frequency to 500 Hz. Reduce input level and note ICS output should be off at 100V and on at 400V.

4.3.17 Transmit audio output, microphone amp. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to position 1, also set MIC switch to position 1 (5 ohms). Connect Pin J1-d to ground. The output level shall be 0.39 volts ± 1 dB across pins J1-N and J1-P. Measure the output voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5. Set microphone switch to position 2, and connect a 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater to pins J1-L and Jl-M. Apply 0.6 volts at 1.0 kHz to the pad, set selector switch to position 1, and jumper pin Jl-d to ground. Observe that the output across pins Jl-N and Jl-P is 0.39 volts ± 1dB. Measure the output voltage at pin pairs JI-S and JI-T, JI-V and JI-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5 to determine compliance.

4.3.17.1 Aux mic input. Set-up equipment as in above paragraph. Apply 30 mV at 1 kHz between J1-G (HI) and J1-c (LO), the output voltage at J1-N, P with the selector at position 1 and J1-d jumpered to ground shall be 0.39V ± 10%.

4.3.17.2 Remote transmit selection.

4.3.17.2.1 Remote select, sequential mode. Apply 0.6 volts at 1.0 kHz to a 5,000 to 5 ohm pad connected to pin J1-L and J1-M and set selector switch to RMT. Connect a resistor of 1K to 10K ohms between a +28 volt source and each of pins J1-f, q, h, r, t. Momentarily remove and then restore +28 volt power to the unit under test. Jumper J1-d to ground (J1-c). Measure audio output voltage at J1-S, T (channel 2). It shall be 0.39 volts ±l dB. Remove jumper at J1-d and momentarily jumper J1-B to ground then restore jumper to J1-d and now voltage at J1-S,T should go to less than 10 mV and voltage of J2-V,W should go to 0.39 volts ±l dB. Repeat above procedure for pin pairs J1-k, s and J1-p, q (channels 4 and 5 and then J1-N,P (channel 1)).

4.3.17.2.1.1 Initialization. With channel select at RMT and the sequential mode selected (J2-h open), the selected channel shall be channel 1 immediately after power turn-on unless channel 1 radio is off or disconnected (28 volts removed from pin J1-f). In that case the first channel with a live radio is selected. If there are no live radios, channel 1 is selected.

4.3.17.2.2 Remote select, SP5T mode. With same setup as paragraph 4.3.17.2.1, move jumper J2-h to ground. Remove the jumper from J1-d to ground. Add a jumper from J2-Z to ground and measure J1-N, P audio output to be .39 volts ± 1.0 dB. Move second jumper to J2-T to ground and measure voltage at J1-S, T. Move second jumper to J1-a to ground and measure

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J1-V, W. Move second jumper to J1-F to ground and measure J1-k, s. Move second jumper to J1-B and measure J1-p, q. Audio output to be .39 ± 10% volts for each channel.

4.3.17.3 Remote indicator. Verify the unit operates from 27.5 volt dc aircraft power and maintains performance over the entire range of 17.0 to 33.0 volts dc. Verify the input power does not exceed 2 watts at 28 volts dc and 2.5 watts at 5 volts ac/dc.

4.3.17.3.1 Weight. Measure the weight of the completely assembled remote indicator unit and verify unit does not exceed 1.7 pounds.

4.3.17.3.2 Outline dimensions. Measure the outline dimensions of the completely assembled unit and demonstrate unit is within the following limitations and exclusive of the cable:

Panel/Housing Height - 1.875 inches maximum.

Panel width - 5.75 inches maximum.

Housing width - 5.0 inches maximum.

Housing depth (from mounting surface and less connector) - 2.9 inches maximum.

4.3.17.3.3 Chromaticity. Verify normal legend illumination is ANVIS GREEN A with a U' coordinate of 0.099, V' coordinate of 0.544 and a radius of 0.034 on the 1976 UCS chromaticity diagram measured at 0.1 foot-lambert display luminance.

4.3.17.3.4 Spectral radiance. Verify the Anvis Radiance does not exceed 1.7×10^{-10} AR when illuminated to produce 0.1 foot-lambert display luminance.

4.3.17.3.5 Chromaticity and spectral radiance. Verify the chromaticity falls within the area noted as aviation green with a dominant wavelength of 510 ± 15 nanometers (nm) and allow no more than one percent of the relative peak amplitude to exist between 600 and 930 nm.

4.3.17.3.6 Input power. Verify the unit operates from normal 27.5 volt dc aircraft power and maintains required performance other than light intensity, over the entire range of 17 to 33 volts dc and when subjected to overvoltage. In normal operation (one annuniciator on) the input power is 28 volts dc (2 watts max) and 5 volts ac/dc (2.5 watts max). During lamp test the input power at 28 volts dc is 6 watts maximum.

4.3.17.3.7 Annuniciator control. Verify when the appropriate control line is grounded the corresponding dual lamp anniciator lights. When a voltage in the range of 17 to 33 volts is applied to the lamp test line all the annunciator lamps will light simultaneously but not at normal full brightness. When a ground is applied to the bright/dim line, the annunciator lamps which are on, will be bright.

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4.3.17.3.8 Transient protection. Verify the units meet specified requirements after subjection to a + 80 volt pulse for 0.2 second duration three times within one minute. The unit also meet specified requirements after subjection to a ± 600 volt spike of 5 µs ± 0.2 µs duration, applied three times within one minute.

4.3.17.4 Piggy back controls. Apply .6V at 1 kHz to the 5,000 to 5 ohm pad connected to pin J1-L and J1-M and set MIC switch to position 1 (5 ohms). Refer to Table IV for front panel control settings and DC inputs to control lines and desired output levels and perform all 24 steps to determine compliance with piggyback operation.

4.3.17.4.1 Interphone operation. When the ICS is keyed by ICS PTT (J1-b to J1-c) or master unit HOT MICA or by master unit VOX or by RAD PTT with ICS selected on master unit, verify the master unit ICS audio circuit is selected and keyed unless the master unit is in remote and the slave unit is in manual with ICS selected.

4.3.17.4.2 Master unit in manual mode. Verify when the master unit is in manual mode, the two unit Station meets all receive mode requirements for all receive channels. Channel selection is by the select switch of the master unit only.

4.3.17.4.3 Slave unit in manual mode. When the slave unit is in manual mode and the master unit is in manual mode, verify operation is in accordance with 3.17.4.2. When the slave unit is in manual mode and the master unit in remote mode, channel selection will be only by means of the slave unit select control. When a transmit channel is not being keyed the station will meet all the receive mode requirements for all receive channels. When either PVT or ICS is selected on the slave unit mic audio will be applied to the selected interphone line corresponding to the slave unit. When the PVT channel or a transmit channel is selected but not keyed and interphone is keyed mic audio will be applied to the ICS line corresponding to the master unit.

4.3.17.4.4 Both units in remote mode. Verify when RMT is selected on both units transmit selection will be by sequential selection only (momentary ground to J1-B). Channel selection will be per paragraph 3.17.2.1, the next channel after channel 5 on any unit will be channel 1 on the other unit.

4.3.18 Audio, headset amplifier (amp).

4.3.18.1 Audio output, single tone, headset amp. Set MIC switch to position 1 (5 ohms). set channel select switch to ICS. Adjust all volume controls to maximum clockwise and push all monitor switches in (ON). Connect a 1.0 kHz input at 2.75 volts to pin pair J2-A,B. The audio output shall be 1.25 volts ± 5% at pin pair J2-D,E across 8 ohms. Apply a 3.9 volt and then a 5.4 volt input signal at 1.0 kHz to determine compliance with limiting threshold and maximum limited

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output requirements of audio output headset amp. Reset input level until output is 1.25 volts RMS and then set MIC switch to position 2 (150 ohms). The audio output shall be 1.25 volts ± 10%. Return MIC switch to position 1 (5 ohms) and set input level to 2.75 volts. Then apply input in turn to following pin pairs:

J2-V,W	J2-X,Y	J2-F,G	J2-n,p	J2-J,K	J2-d,e	J2-q,r
J2-U,j	J2-k,m	J2-a,b	J2-c,H	J2-s,t	J2-L.M	J2-N,f

The audio output shall be 1.25 volts ± 5% for each of above inputs. Apply 0.39 volts RMS to pin pair J2-R,S to obtain same output and then set channel select switch to PVT and apply 0.39 volts RMS to pin pair J2-g,P to obtain same 1.25 volts ± 5% output.

4.3.18.2 Audio output, 2-tone test, headset amp. Apply a 5.4 volt input signal of 1.0 kHz to a selected input and a 2.75 volt, 1.5 kHz signal to a selected non related input. Observe that the 1.5 kHz output signal is not reduced in amplitude by greater than 9 dB by the presence of the 1.0 kHz signal to determine compliance.

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Table IV. Piggy back control test settings.

INPUT (NOTE 1)						OUTPUT			
	RAD 1	RAD 2	RAD 3	RAD 4	RAD 5	PTT RAD	PTT PVT	SEL CTRL	XMT 1
ITEM	J2	J2	J1	J1	J1	J1	J1	-	J1
	T	Z	a	F	B	d	A		f
1	HI	HI	LO	LO	HI	LO	HI	PVT	HI
2						HI			
3		LO	HI						
4				HI					
5							LO		
6							HI	ICS	
7				LO					
8		HI	LO						
9								1	
10		LO	HI						
11				HI					
12						LO			LO
13				LO					HI
14			LO						LO
15				HI	NOTE	HI		RMT	HI
16		HI		LO	HI				
17	LO								
18	HI	LO							
19	LO	HI		HI					
20	HI					LO			LO
21		LO	HI		NOTE	HI			HI
22					HI	LO			LO
23				LO					HI
24				HI		HI			

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Table IV. Piggy back control test settings - Continued.

INPUT (NOTE 1)						OUTPUT			
	STAT PVT	STAT ICS	STAT 1	STAT 5	SER OUT	PVT AUD	ICS AUD	XMT AUD	NOTES
ITEM	J1 X	J1 Y	J1 Z	J1 n	J2 i	J2 g,P	J2 R,S	J1 N,P	
1	LO	HI	HI	HI	LO	YES	NO	NO	
2						NO	YES		
3	HI						NO		
4	LO						YES		
5						YES	NO		
6	HI	LO				NO	YES		
7		HI					NO		
8		LO					YES		
9		HI	LO						
10			HI				NO		
11			LO				YES		
12							NO	YES	
13			HI					NO	
14			LO					YES	
15			NOTE				YES	NO	NOTE2
16			HI		HI				
17							NO		
18									
19			LO		LO		YES		
20							NO	YES	
21							YES	NO	NOTE2
22							NO	YES	
23			HI		HI			NO	
24			LO		NOTE		YES		NOTE3

Note 1: Pin J2-h, J1-b are HI and ICS SW at HOT MIC.

Note 2: In this mode, momentarily grounding J1-B will select next channel, repeat until channel 1 is selected.

Note 3: In this mode after channel 5 is selected the next time J1-B is momentarily grounded then J2-i will momentarily go HI for about 50 MS and then go LO again as channel 1 is selected.

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4.3.19 Response, headset amp. Set MIC switch to position 1 (5 ohms) and apply a 1.0 kHz input at 0.39 volts to pins J2-R and J2-S and adjust volume control for a 0.7 volt output across pins J2-D and J2-E and using this output level as a reference level while maintaining a constant input level, measure the frequency response from 350 to 4,500 Hz and 87 Hz and 18,000 Hz to determine compliance.

4.3.20 Distortion headset amp.

4.3.20.1 Distortion. With MIC switch in position 1, volume control fully CW and all monitor controls in (ON) and fully clockwise apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio audio input in turn and measure the headset amp output distortion at 350 Hz for compliance.

4.3.20.2 Distortion full range headset amp. Apply 0.39 volts to ICS and PVT audio input and 2.75 volts to each radio channel audio input in turn to obtain 1.25 volts ± 5% output at 1.0 kHz across pins J2-D and J2-E. Measure the headset amp output distortion at 350 Hz, 1000 HZ and 4500 Hz for compliance. Reduce the input voltage to .055 volts for ICS and PVT audio inputs and to 0.39 volts for each radio channel audio input and measure the zero crossing distortion to determine compliance.

4.3.21 Broadband noise headset. Set the selector switch to ICS position and all received audio monitor switches to the ON position. With no received audio, ICS or PVT, input signals, measure the broadband noise across pins J2-D and J2-E to determine compliance.

4.3.22 Isolation, monitor switches/audio lines. Set selector switch to position 1, all monitor switches out (OFF) and maximum CW, HOT MIC switch to NORM and MASTER volume to maximum CW. Apply a 1.0 kHz signal at 2.75 volts to pin pair J2-A,B to obtain 1.25V ± 5% output across J2-D,E. Set selector switch to ICS position and measure the output level relative to the 1.25 volt output to determine compliance. In turn, apply the input to pin pairs J2-V,W and J2-U, j and J2-X,Y and J2-k,m and J2-F,G and J2-a,b and J2-n,p, and J2-c,H and J2-J,K and J-2-s,t. As each input is applied, measure the output level relative to the 1.25 volt output to determine compliance. Set selector switch to position 2, apply above input to J2-V,W to obtain 1.25V ± 5% output. Set selector switch to position 3, apply above input to J2-U,j to obtain 1.25V ± 5% output. Set selector switch to position 4, apply above input to J2-X,Y to obtain 1.25V ± 5% output. Set selector switch to position 5 apply above input to J2-k,m to obtain 1.25V ± 5% output. Leave selector switch; in position 5 and apply 0.39 Volt signal input to pin pair J2-R,S to obtain 1.25V + 5% output. Set Hot MIC switch to ICS OFF and measure the output relative to 1.25V to determine compliance. Return Hot MIC switch to NORM and push all monitor switches in (ON) and apply 2.75V, 1.0 kHz to a radio input pair and measure the output level at the other radio input pin pairs. Repeat these measurements until all audio lines have been tested to determine compliance.

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4.3.23 Isolation, transmitter lines. Apply 0.39 volts at 1.0 kHz to pin pair J2-R and J2-S and obtain 1.25 volts at J2-D, E. Set MIC switch to position 1 (5 ohms) and selector switch to positions 1 through 5 and measure crosstalk, with and without pin J1-d jumpered to ground, at pin pairs J1-N and J1-P, J1-S and J1-T, J1-V and J1-W, J1-K and J1-s, and J1-p and J1-q respectively, while applying in turn 0.39 volts to ICS and PVT inputs and 2.75 volts to all monitor inputs except that corresponding to the selected channel. Also measure at pins J1-f, J1-g, J1-h. J1-r and J1-t, terminated in 270 ohms to chassis ground, as selector switch is set to positions 1 through 5, respectively, and pin J1-d is jumpered to ground for each input. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M. Set MIC switch to position 1 and selector switch to position ICS, PVT and 1 through 5 and measure crosstalk on all transmit audio pairs, except that corresponding to the selected channel, with the appropriate key line jumpered to ground. Now set MIC switch to position 2 and measure crosstalk on all unselected transmit audio pairs at each position of the selector switch while applying 0.6 volts at 1.0 kHz to a 2,400 ohm to 150 ohms pad connected to pins J1-L and J1-M with the appropriate key line jumpered to ground. The crosstalk levels at the transmitter audio and transmitter control lines shall comply with specified requirement.

4.3.24 Transient protection. Under standard test conditions, apply a +80 pulse for 0.2 second duration between pins J1-C and J1-D. Apply each pulse three (3) times within one (1) minute. Perform both of these tests to determine compliance. In addition, apply at least three (3) aperiodic spike waveforms between Pins J1-C and J1-D. The spike waveform, produced by a generator with a source impedance of 50 ± 50 ohms shall satisfy the following requirements:

Open-circuit voltage: +600 volts peak
Risetime: 0.9 ± 0.2 microseconds
Falltime: 10.0 ± 1.0 microseconds

4.3.25 Headset amp unterminated. With input power applied and input signals removed, disconnect the 8 ohm load resistor across pins J2-D and J2-E. Monitor the amplifier output each time the switches and volume controls are operated. Short pins J2-D and J2-E to ground for 1 minute. Observe that no unusual emanations occur from the unit while shorted and that no damage is sustained when the short is removed, to determine compliance.

4.3.26 Transmitter control.

4.3.26.1 Transmitter control 1 amp. Jumper pins J1-c and J1-d set selector switch to position-1, and apply a plus dc voltage through a current limiting device to pin J1-f (negative to pin J1-D) to obtain a 1.0-ampere current flow. In turn, set selector switch to positions 2 through 5 as the dc voltage is applied to pins J1-g, J1-h, J1-r and J1-t. These current measurements shall be made to determine compliance.

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4.3.26.2 Transmitter control, GO/NO GO. Follow procedure in paragraph above except a GO/NO GO test may be made using a low current source. Connect a low current +28 volt source separately to all of pins J1-f, g, h, r, t and place selector switch in RMT. Remove jumper from J1-d to ground. Momentarily ground pin J1-B and note that there is automatic selection of the next radio channel as evidenced by voltage at next pin going to less than 1 volt when J1-d is grounded. Momentarily ground J1-B enough times to scan through all 5 channels. Remove power from J1-t and then momentarily ground J1-B sufficient times to note that the next channel selected after channel 4 is channel 1. Continue to remove power from each pin in turn noting that the remote selection procedure skips all channels of which the control pins are not powered. When power is removed from all pins channel 1 will be selected. When power is removed from the channel 1 pin and power is applied to the channel 3 pin (or any other pin) the channel corresponding to the powered pin is selected.

Now jumper J2-h to ground and ground J2-Z momentarily. Note that channel 1 is selected only when J2-Z is grounded. Then ground J2-T and note selection and keying of channel 2. Similarly test channels 3, 4, 5 by momentarily grounding J1-a, F, B respectively.

4.3.27 Function selector logic.

4.3.27.1 Function selector logic, .5 amp. Jumper pins J1-c and J1-A, set selector switch to PVT, and apply a plus dc voltage through a current limiting resistor to pin J1-z (negative to pin J1-D) to obtain a 0.5-ampere current flow. Jumper pins J1-c and J1-d. In turn, set selector switch to positions ICS, and 1 through 5 as the dc voltage is applied to pins J1-Y, J1-Z, J1-j, J1-m and J1-n. These current measurements shall be made to determine compliance.

4.3.27.2 Function selector logic, GO/NO GO. Follow procedure above except a GO/NO GO test may be made using a low current source. Connect a low current +28 volt source separately to all of pins J1-f, q, h, r, t and place selector switch in RMT. Momentarily ground pin J1-B and note that there is an automatic selection of the next channel as evidenced by voltage at the next of pins J1-Z, i, j, m, n going to less than 1 volt. Momentarily ground J1-B sufficient times to scan through all 5 channels to verify proper operation.

Now Jumper J2-h to ground and momentarily ground J2-Z, T, J1-a, F, B in turn to see that channels 1 through 5 are selected respectively as evidenced by voltage at J1-Z, i, j, m, n going to less than 1 volt respectively only when the corresponding control is momentarily grounded.

4.3.28 Microphone power. Set microphone selection switch in position 2, attach a 1000 ohm resistive load to pins J1-L and J1-M. Measure the dc voltage across the resistive load to determine compliance.

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4.3.29 Received audio input attenuation during transmit. Set selector switch to position 1, all monitor switches to ON and fully clockwise position, and HOT MIKE switch to NORM. Apply a 0.39 volt signal input at 1.0 kHz to pin pair J2-q and J2-P. Obtain 1.25 volts±10% at pin pair J2-D,E. Apply a ground to pin J1-d and measure the attenuation of the headset amplifier output at pins J2-D and J2-E with respect to the normal 1.25 volt output. In turn, apply the proper input, (0.39 volts to J2-R and J2-S, J2-q and J2-P) (2.75 volts to all other audio inputs), to all audio input pairs. Measure the headset amplifier output attenuation, apply a ground to pin J1-d and repeat this test for selector switch positions 1 through 5 described in Table II to determine compliance. Connect a jumper between pin J1-F and J1-c (key COMM). Perform tests again. This time 12 dB attenuation should occur even on selected channels.

4.3.30 Audio level (volume adjustments).

4.3.30.1 Main volume control. Connect a 1.0 kHz input signal successively to each controlled received audio Input (i.e., PVT, ICS, 1, 2, 3, 4, 5, NAV 1, NAV 2, DIR) and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor (where applicable) and master audio level controls set for maximum audio output. Adjust the main volume control for minimum audio output and measure this voltage for compliance.

4.3.30.2 Monitor audio level adjustment. Connect a 1.0 kHz input signal successively to each switched received audio input (i.e., 1, 2, 3, 4, 5, NAV 1, NAV 2), and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor and master audio level controls set for maximum audio output. Adjust the appropriate individual audio level control for minimum audio output and measure this voltage for compliance.

4.3.31 TEMPEST testing. The equipment shall be tested in accordance with the procedures of NACSEM 5100 and the contractor prepared, Government approved TEMPEST Test Plan.

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4.3.32 Electromagnetic interference. Verify that electromagnetic emissions in the frequency of 30 Hz to 50 KHz do not appear on dc power leads in excess of the values shown in Figure 1. Verify that electromagnetic emissions in the frequency of 10 KHz to 50 MHz do not appear on ac or dc power leads in excess of Figure 1. Verify that electromagnetic emissions in the frequency of 30 Hz to 50 MHz do not appear on control leads, signal; leads, and interconnecting cables in excess of the values shown in Figure 1. Verify that electromagnetic emissions in the frequency of 10 KHz to 50 MHz do not appear on ac or dc power leads in excess of the values shown in Figure 2. Verify that electromagnetic emissions in the frequency of 30 Hz to 50 MHz do not appear on control leads, signal; leads, and interconnecting cables in excess of the values shown in Figure 2. Narrowband field emissions in the frequency range of 14 KHz to 1,000 MHz shall not be generated and radiated in excess of the values shown in Figure 3 at a test distance of 3 meters on each face of the equipment. Broadband field emissions in the frequency range of 14 KHz to 10 GHz shall not be generated and radiated in excess of the values shown in Figure 4 at a test distance of 3 meters on each face of the equipment. Horizontally and vertically polarized waves in the frequency range of 30 MHz to 10 GHz shall meet the same limit. Radiated switching transients resulting from automatic cycling of electronic switching circuitry resulting from keying of transmitters shall not be generated in excess of the values shown in Figure 4 at a test distance of 3 meters on each face of the equipment. Radiated switching transients resulting from manually controlled operational mode switching functions shall not exceed the values shown in Figure 5 at a test distance of 3 meters. The unit(s) controls shall be adjusted as defined except as may be required to simulate special conditions. Selector switch to ICS, Hot MIC switch to Hot MIC, all monitor switches in ON, and MIC switch to position 1 (5 ohms), volume controls full clockwise position. The length of interconnecting cable shall be approximately six (6) feet. All emission tests shall be conducted with maximum input/output signal processed through the audio channels.

4.3.32.1 Audio susceptibility. The equipment shall not be degraded in the frequency range of 30 Hz to 50 KHz when subjected to electromagnetic energy injected on its dc power leads. The unit headset (J2-D and J2-E) output shall not exceed 4.0 mV between 30 Hz and 50 KHz. The interphone (J2-R and J2-S) output shall not exceed 13 mV between 30 Hz and 50 KHz. The interphone output J2-R/ J2-S shall be monitored with a scope and any deviation from normal output ±3 dB shall be recorded. In addition to the above, the noise output on headset and interphone lines shall be measured. These tests and measurements shall be made to determine compliance with EMI requirements.

4.3.32.2 Conducted susceptibility. The equipment shall not be degraded in the frequency range of 50 KHz 400 MHz. The success criteria for this requirement shall be a maximum noise output for the Headset Amplifier (J2-D and J2-E) as follows: 4.0 mV between 150 kHz and 199 MHz, and between 251 MHz and 400 MHz; 4.0 mV between 200 MHz and 250 MHz. The maximum noise output for the Interphone Terminals (J2-R and J2-S) shall be 13.0 mV.

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4.3.32.3 Radiated susceptibility. No malfunction or degradation of performance shall occur when the equipment is subjected to the radiated fields as follows:

Susceptibility Levels	10 KHz to 400 MHz	Sheltered	Non-Sheltered
	10 KHz - 1.9 MHz	10 V/M	1 V/M
	2.0 MHz - 29.99 MHz	20 V/M	5 V/M
	30 MHz - 400 MHz	50 V/M	10 V/M

The success criteria for this requirement shall be a Maximum noise output of 4.0 MV for the Headset Amplifier (J2-D and J2-E) and 13.0 mV at the Interphone Terminals (J2-R and J2-S).

The susceptibility levels shall be met for both horizontally and vertically polarized fields from 30 MHz to 400 MHz. The equipment shall be operated and configured for field use during all phases of EMI testing. The contractor shall select the test method subject to review and approval by the government prior to the start of any test.

4.3.32.4 Radiated susceptibility (induction field). Set the equipment to be tested six (6) inches directly above a wire which shall be placed parallel to the front of the unit and 1.25 inches behind the front panel. Connect the wire to a 400 Hz power source and adjust the current to 5 amps, rms. Set the unit controls as in output interphone and sidetone verification test paragraph and measure the output levels to determine compliance.

4.3.33 Bonds, grounds, connectors and cables. Verify bonding for the equipment current path return, RF potentials and shock hazard are so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Visually inspect surface preparation for bonds and grounds for anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Verify bonding impedance does not exceed 2.5 milliohms. Verify case and chassis mounted connectors have approved RF conductive finish and are installed as above. Demonstrate bonding impedance of the connector shell to the case does not exceed 2.5 milliohms. Demonstrate interconnecting cables requiring shields have a minimum of 40 dB shield attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. Visually inspect the shield is constructed of tinned copper braid connectors with cables having shields have an approved RF conductive finish and provided with EMI backshells for terminating the shields.

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4.3.34 Operational inspections. Each unit(s) shall be connected to form an operational system. A talk test shall be performed to determine compliance with 3.34. Each unit(s) shall be connected with at least nine other units (C-11746(V)( )/ARC) to form an operational system. (Substitution of simulated additional units is permissible when operational conditions are accurately duplicated. The substituted fixture shall be approved by the procuring activity.) A talk test shall be performed to determine compliance with 3.19.

4.3.35 Interchangeability. Verify like units, assemblies, subassemblies, modules, and replaceable parts are functionally and physically interchangeable without modification and degradation of performance. Verify that these like units, assemblies, subassemblies, modules, and replaceable parts also are functionally and physically interchangeable with like units of the "standard" C-11746(V)/ARC Communications System Control Unit. The contractor shall select the method of test subject to review and approval by the government prior to the start of any test.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES.

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. Control Unit, Communication System C-11746(V)( )/ARC is intended for use by the pilot, copilot or crew to provide control of radio equipments for voice transmission and reception, direction finding equipment and intercommunication when installed in aircraft and will provide remote transmit selection. The Control Unit, Communication System C-11746(V)( )/ ARC is designed to process audio output to an individual headset from up to five communications receivers and six navigation receivers. When remote transmit selection is to be used the Indicator, Remote ID-2403/ARC may be installed to display the channel selection.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification and any amendment.

b. When a first article is required (see 3.1, 4.3.1, and 6.3).

c. Sample plans for verification inspection and test. For general information only; Sampling in past procurements using a detailed design specification was conducted using MIL-STD-105. MIL-STD-105 is canceled and replaced by American National Standard, ANSI/ASQC Z1.4

d. What conformance verification is required (see 4.3.2 and 6.4).

e. Packaging requirements (see 5.1).

f. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: The C-10716/TRC(V)2. The contractor should apply for nomenclature according to the applicable contract clause.

g. Government Furnished Equipment provided for testing.

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, criticality of test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

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6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include; contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from Table I and their designated frequency based on a risk assessment for the procurement.

6.5 Location of operational inspection and air-seal test. It is desirable that the operational inspection and air-seal test be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled immediately prior to packaging.

6.6 Subject term (key word) listing.

Chromaticity and spectrum
Piggy back configuration
VOX

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian
Army - CR

Preparing Activity
Army-CR
(Project 5895-A472)

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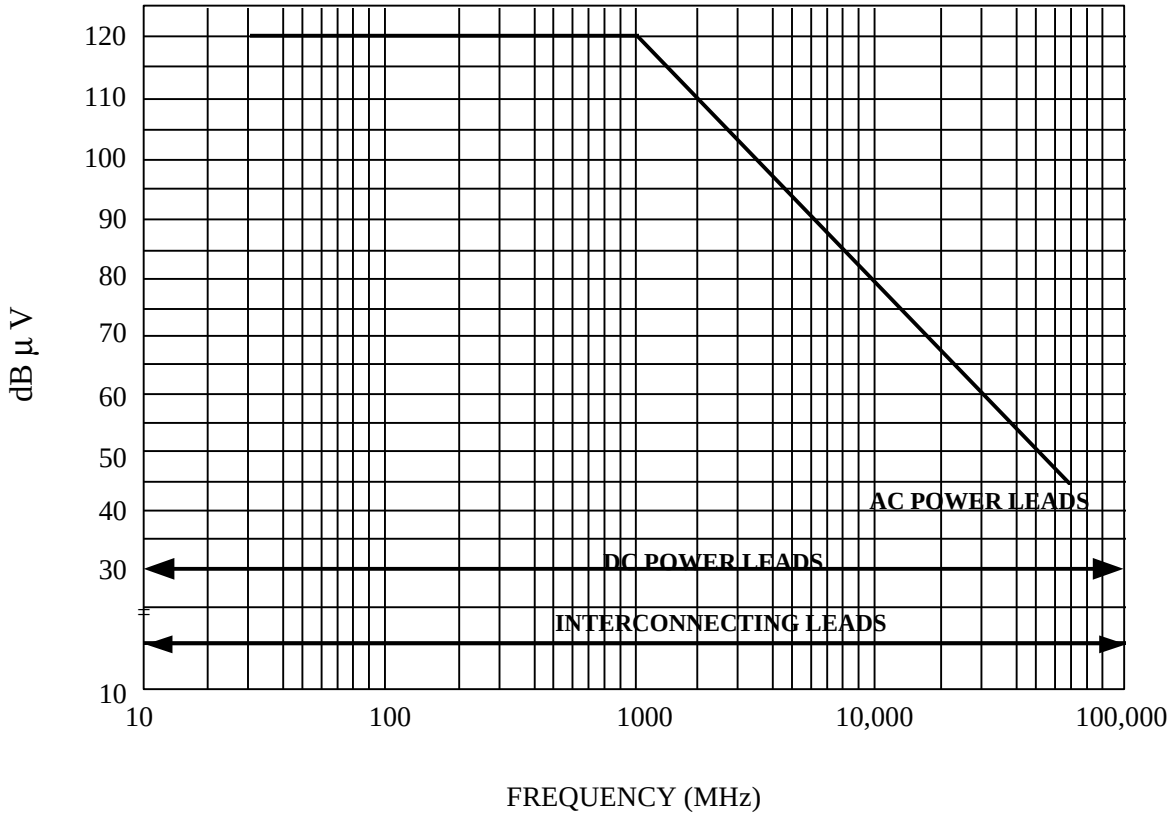


FIGURE 1. Limit for conducted narrowband emissions.

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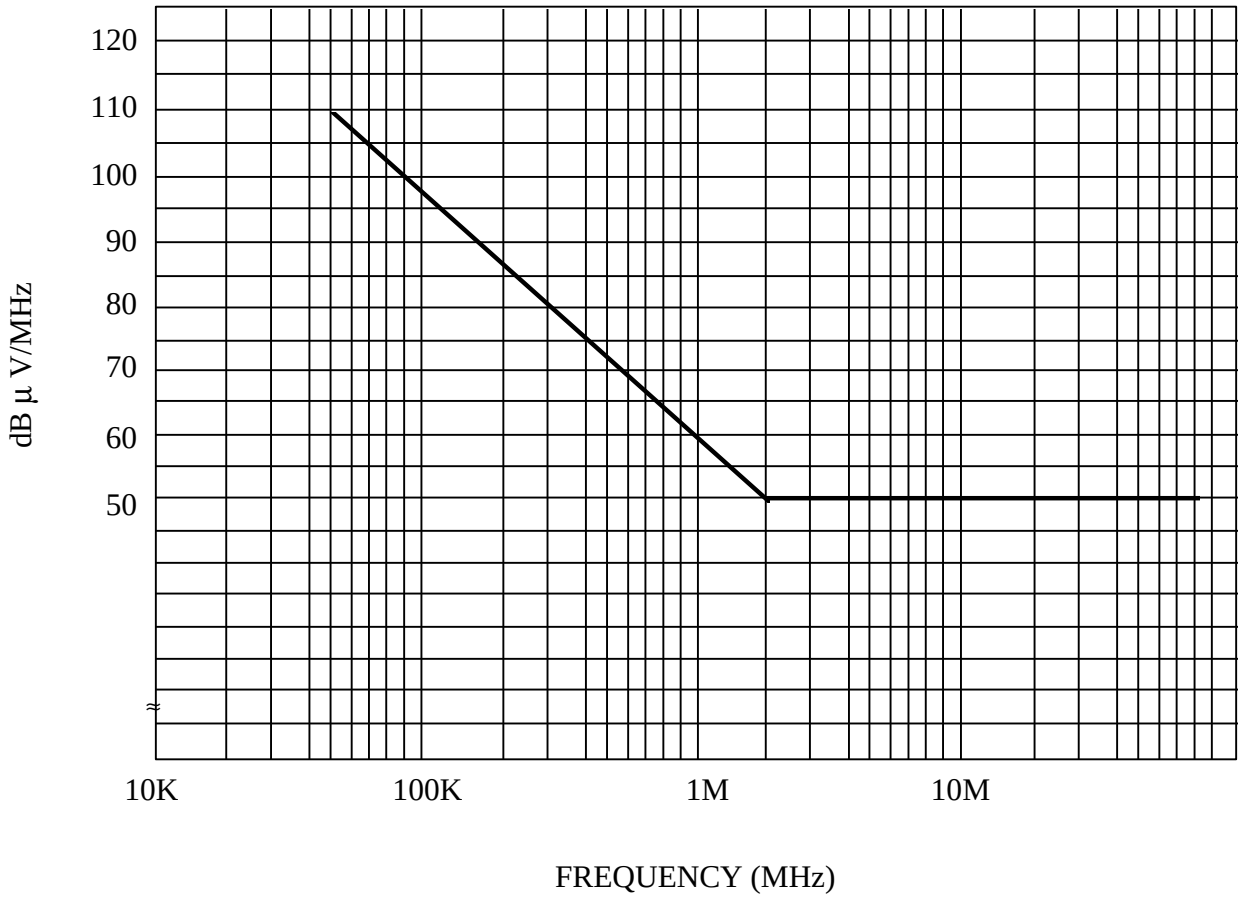


FIGURE 2. Limit for conducted broadband emissions.

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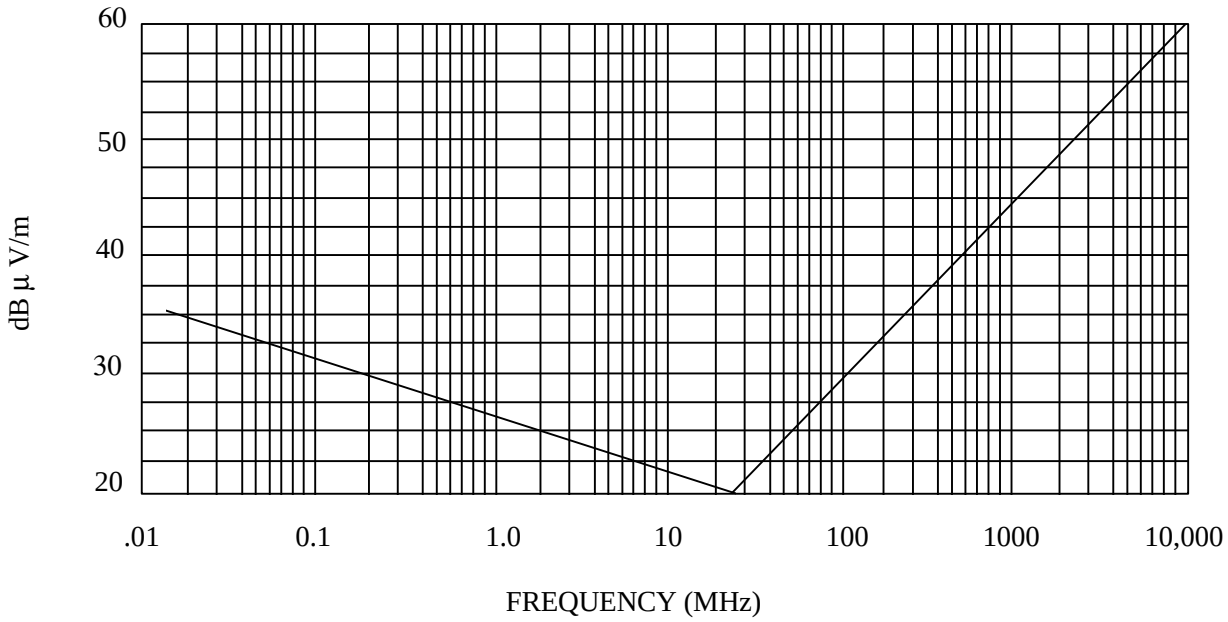


FIGURE 3. Limit for radiated emission narrowband.

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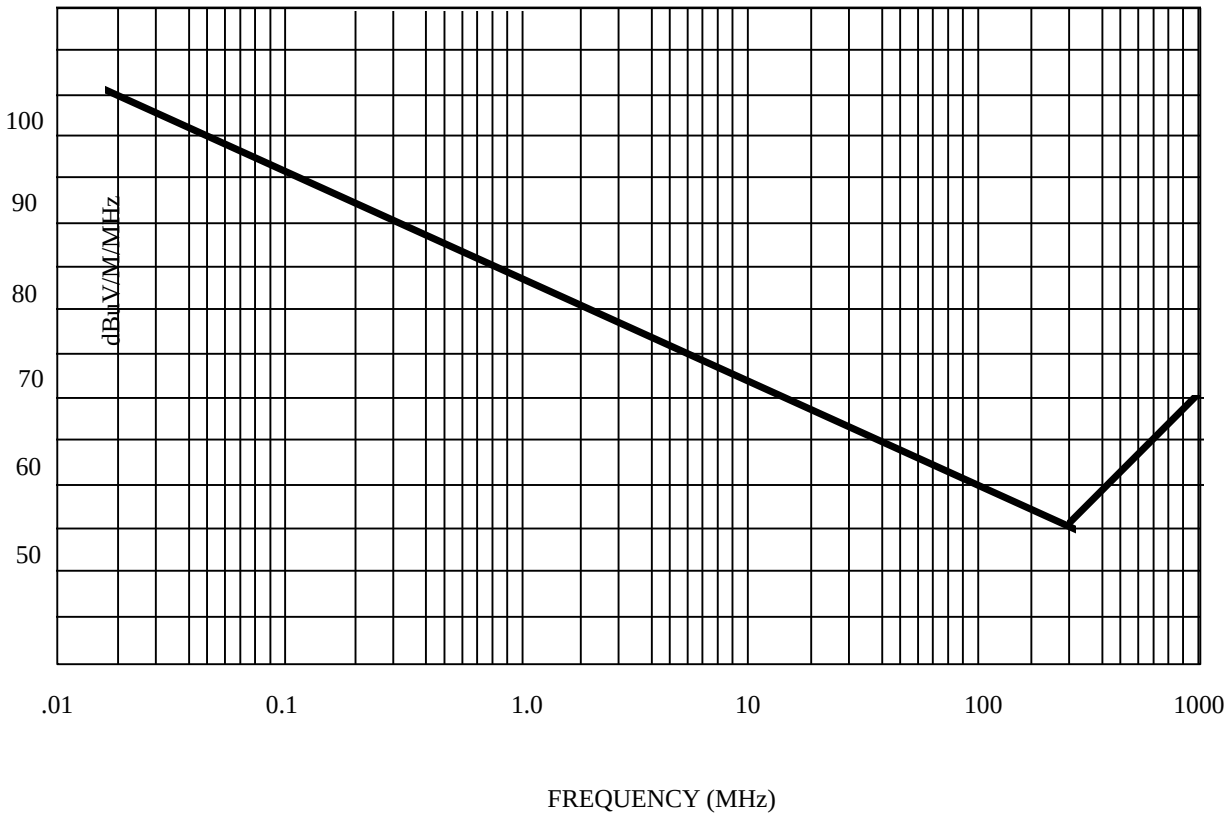


FIGURE 4. Limit for radiated emissions broadband emissions.

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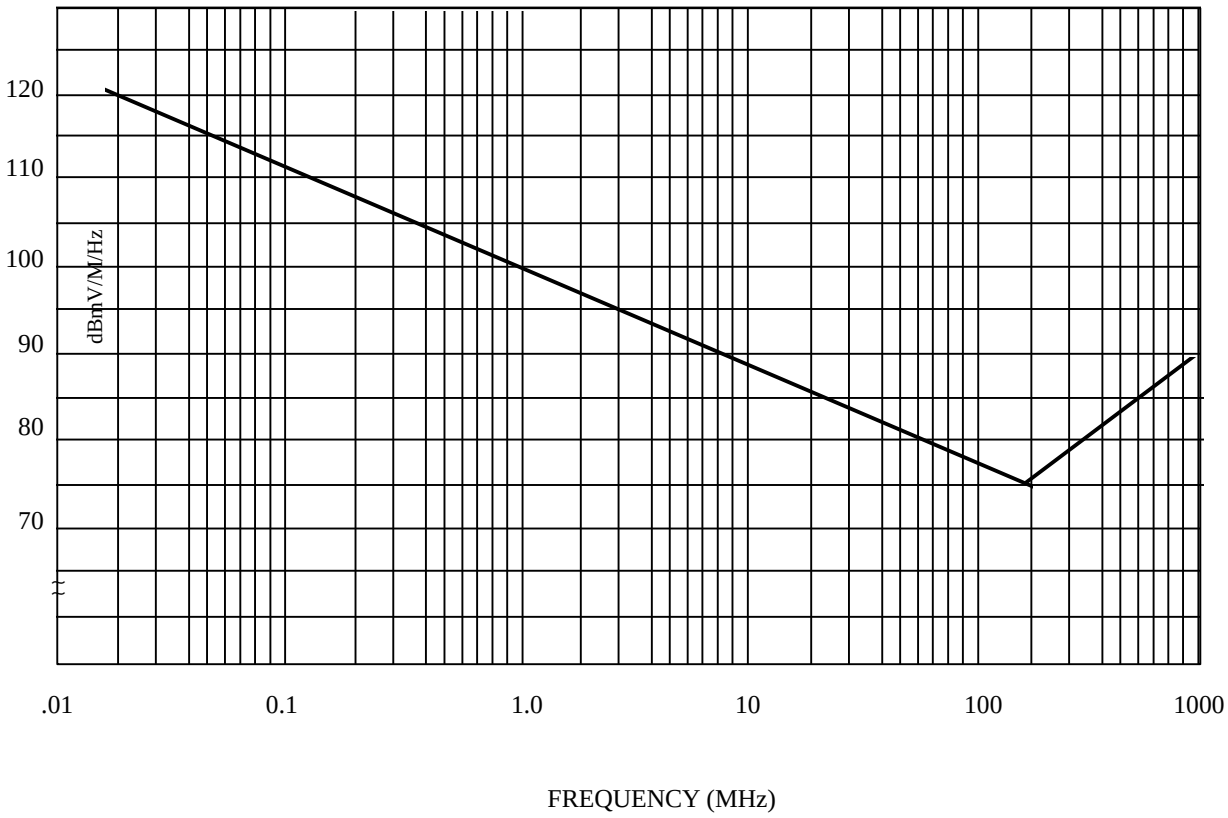


FIGURE 5. Limit for radiated switching transients.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-58111A(CR)	**2. DOCUMENT DATE (YYMMDD)** 971009

3. DOCUMENT TITLE **CONTROL UNIT,COMMUNICATION SYSTEM C-11746(V)( )/ARC**

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732)427-3450 (2) AUTOVON 987-3450
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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